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(54) **PROSTHETIC VALVE WITH ARMS AND FLANGES**

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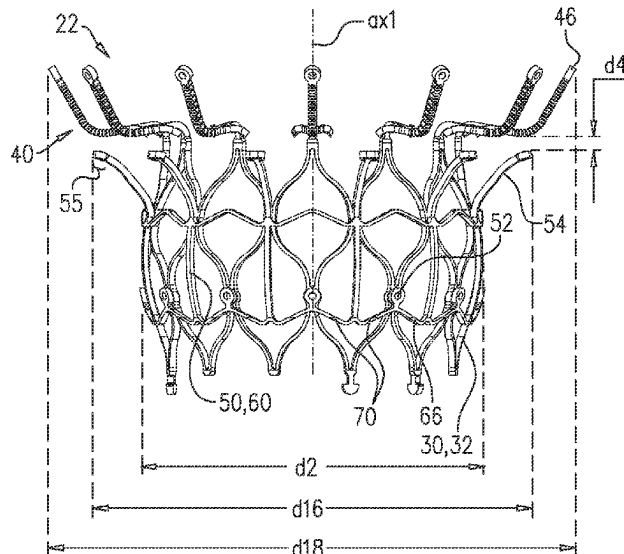
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(57) **ABSTRACT**

A valve frame includes a tubular portion defining a plurality of valve-frame coupling elements disposed circumferentially around a longitudinal axis of the valve frame, and a plurality of arms, extending radially outward from an upstream portion of the tubular portion to define an arm span, and configured to engage tissue in the atrium. An outer frame is coupled to the valve frame, and includes a ring circumscribing the tubular portion, and a plurality of flanges that extend radially outward from the ring to define a flange span, and are configured to engage ventricular tissue. Each arm defines a rigid portion and a flexible portion that is disposed radially outward from the rigid portion, and is more flexible than the rigid portion. The majority of the flexible portion is disposed further radially outward than the flange span. Other embodiments are also described.

9 Claims, 17 Drawing Sheets



Related U.S. Application Data

continuation of application No. 15/668,659, filed on Aug. 3, 2017, now Pat. No. 10,918,481, which is a continuation-in-part of application No. 15/541,783, filed as application No. PCT/IL2016/050125 on Feb. 3, 2016, now Pat. No. 9,974,651.

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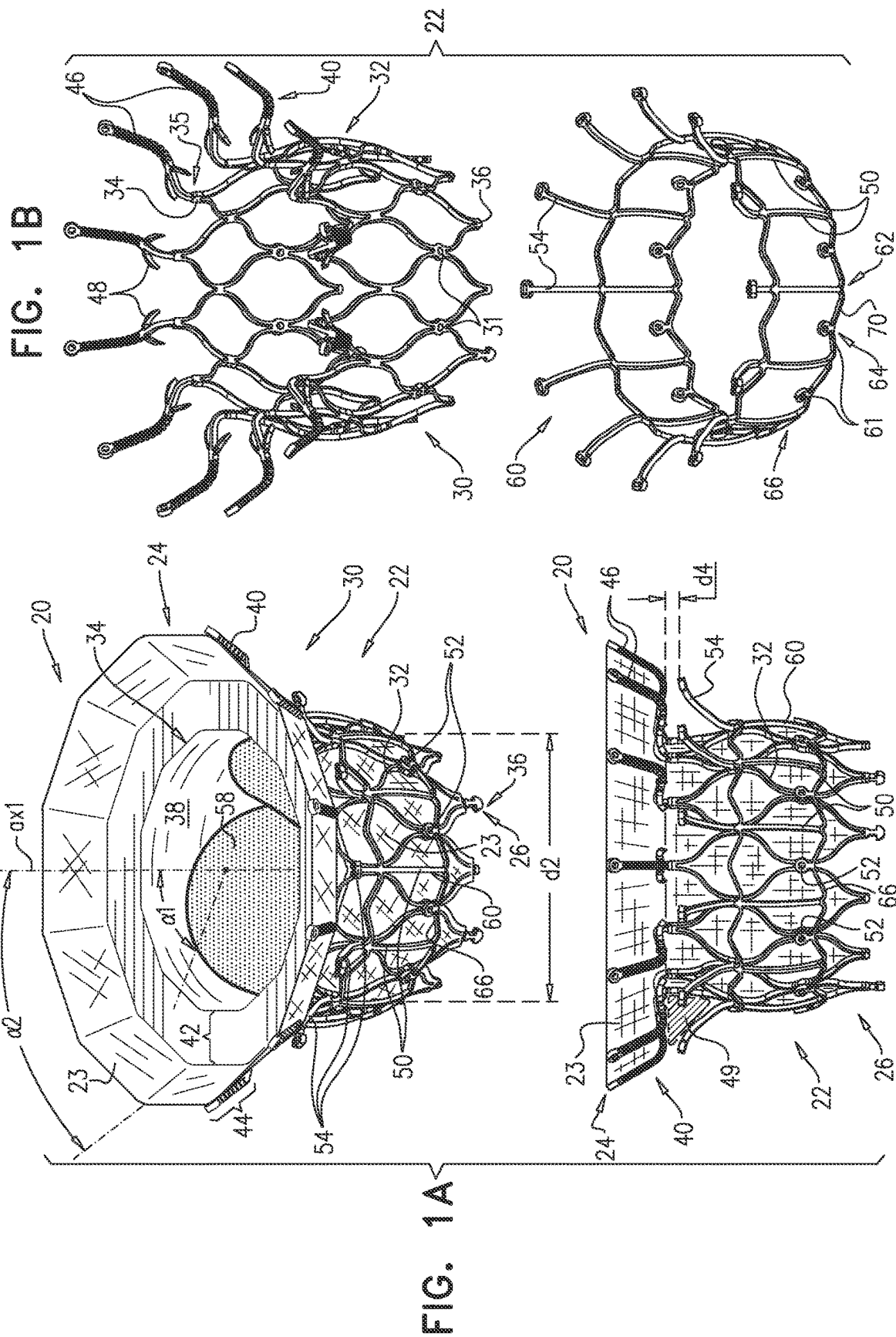


FIG. 2A

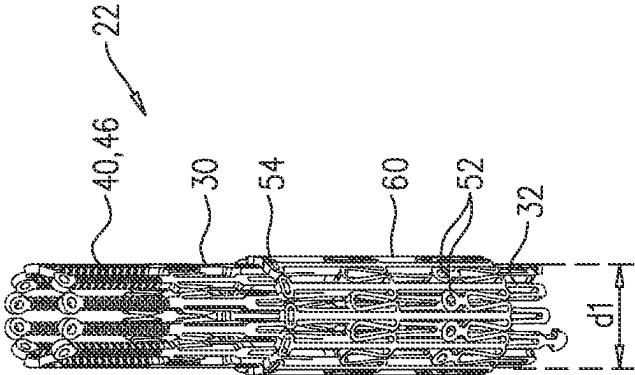


FIG. 2B

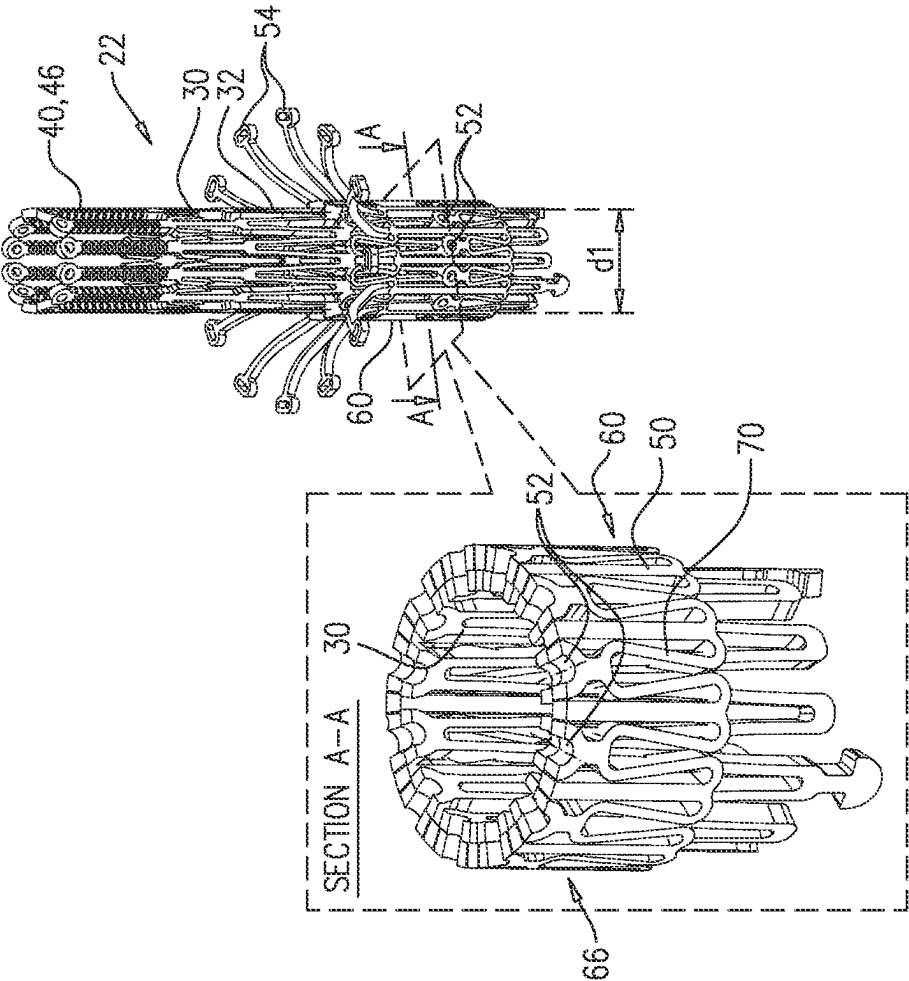


FIG. 2C

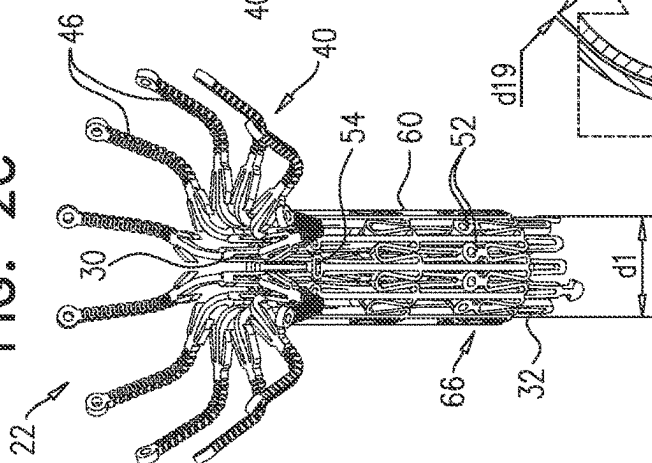


FIG. 2D

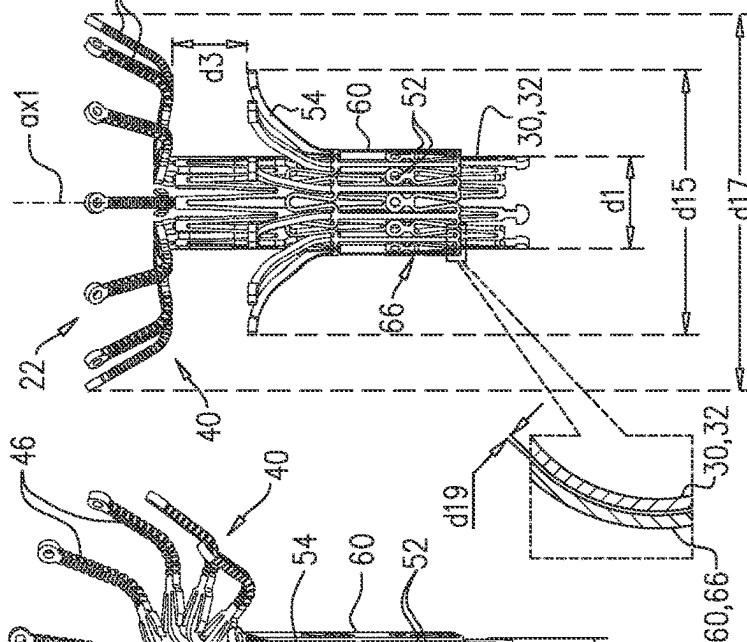
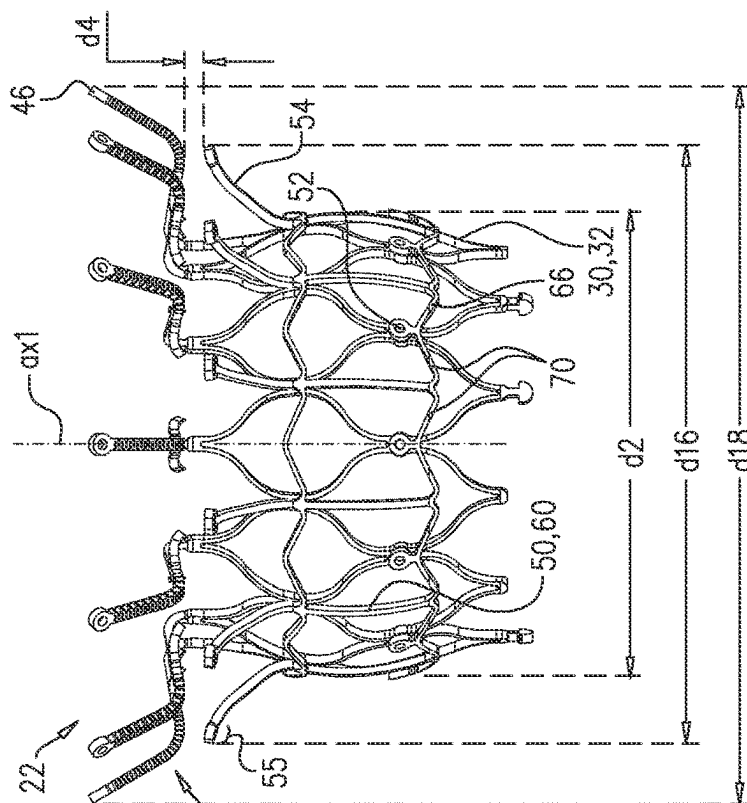


FIG. 2E



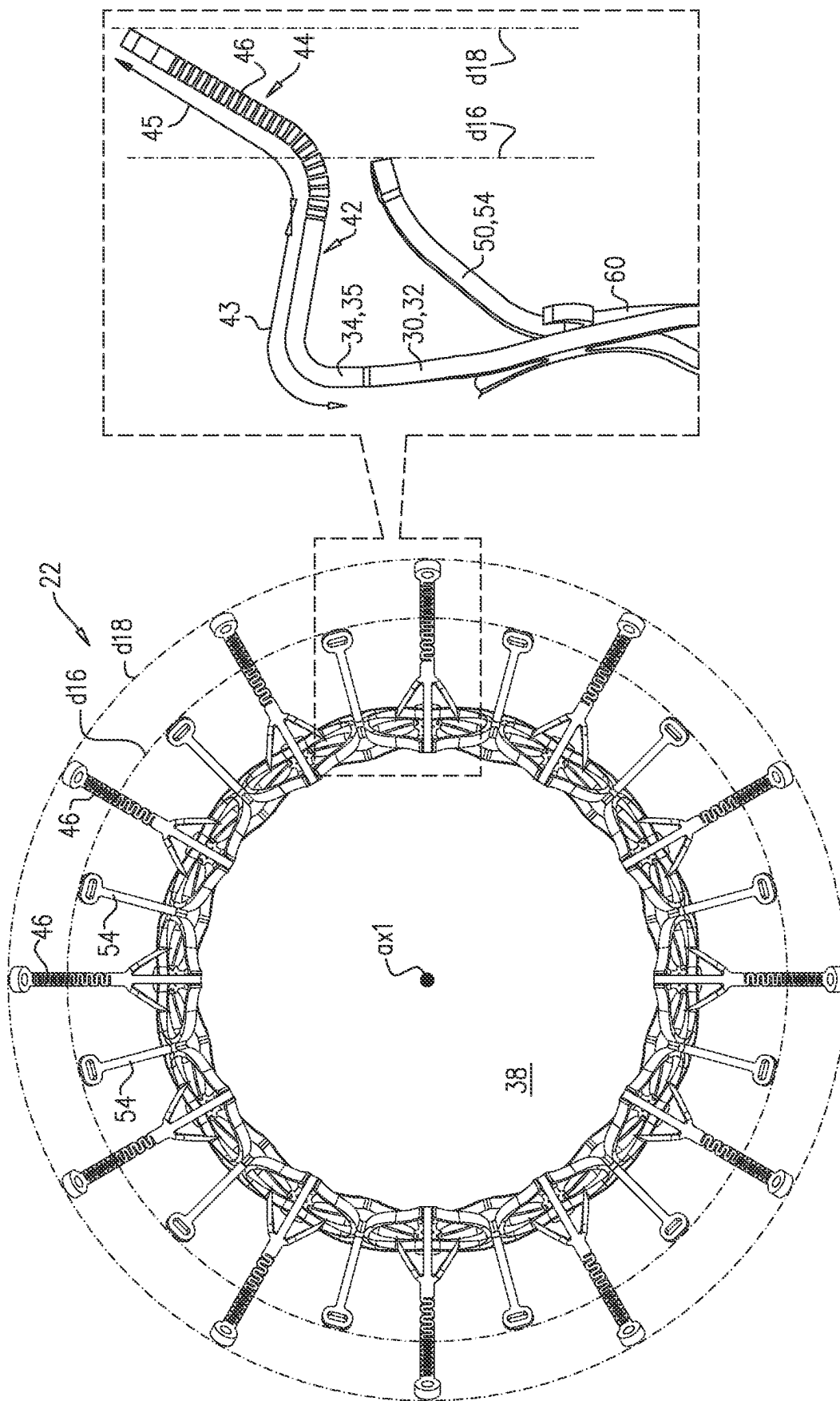


FIG. 2F

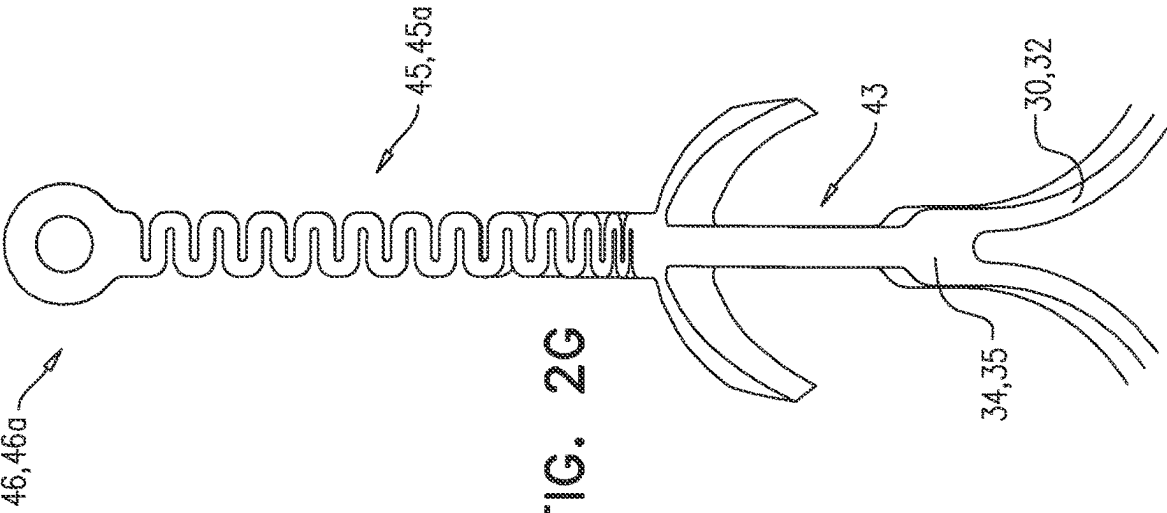
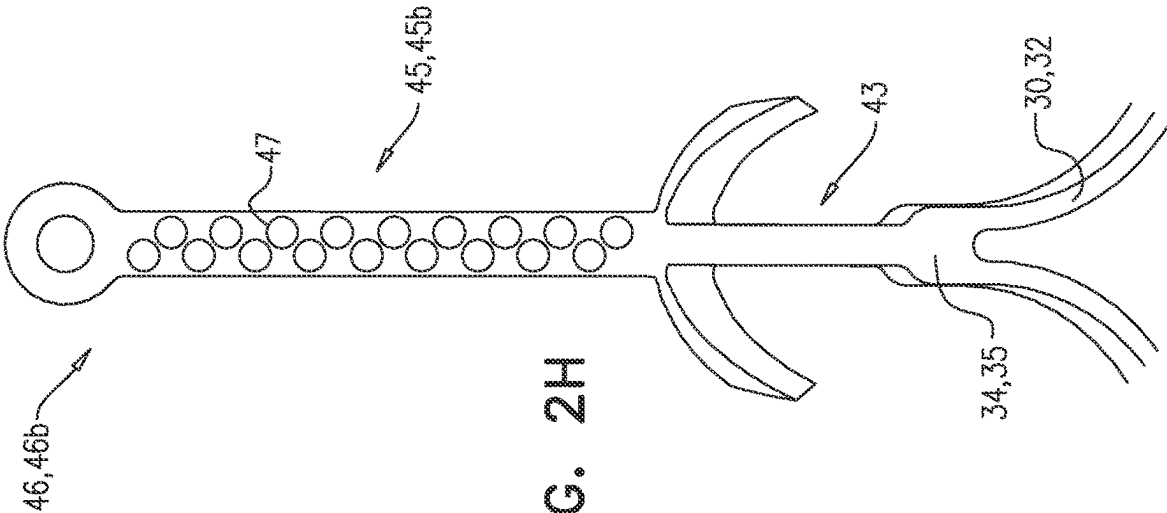


FIG. 3A

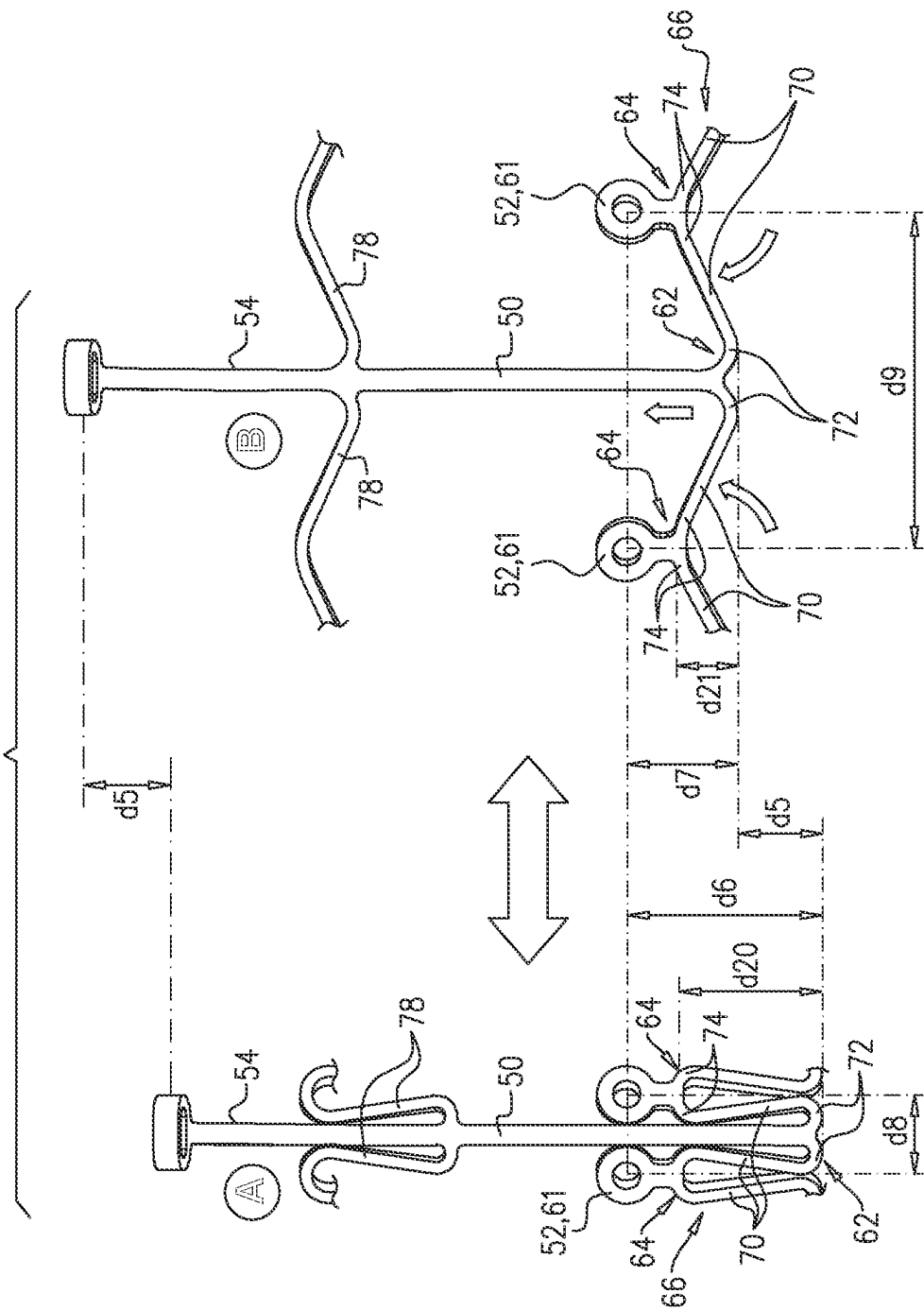


FIG. 3B

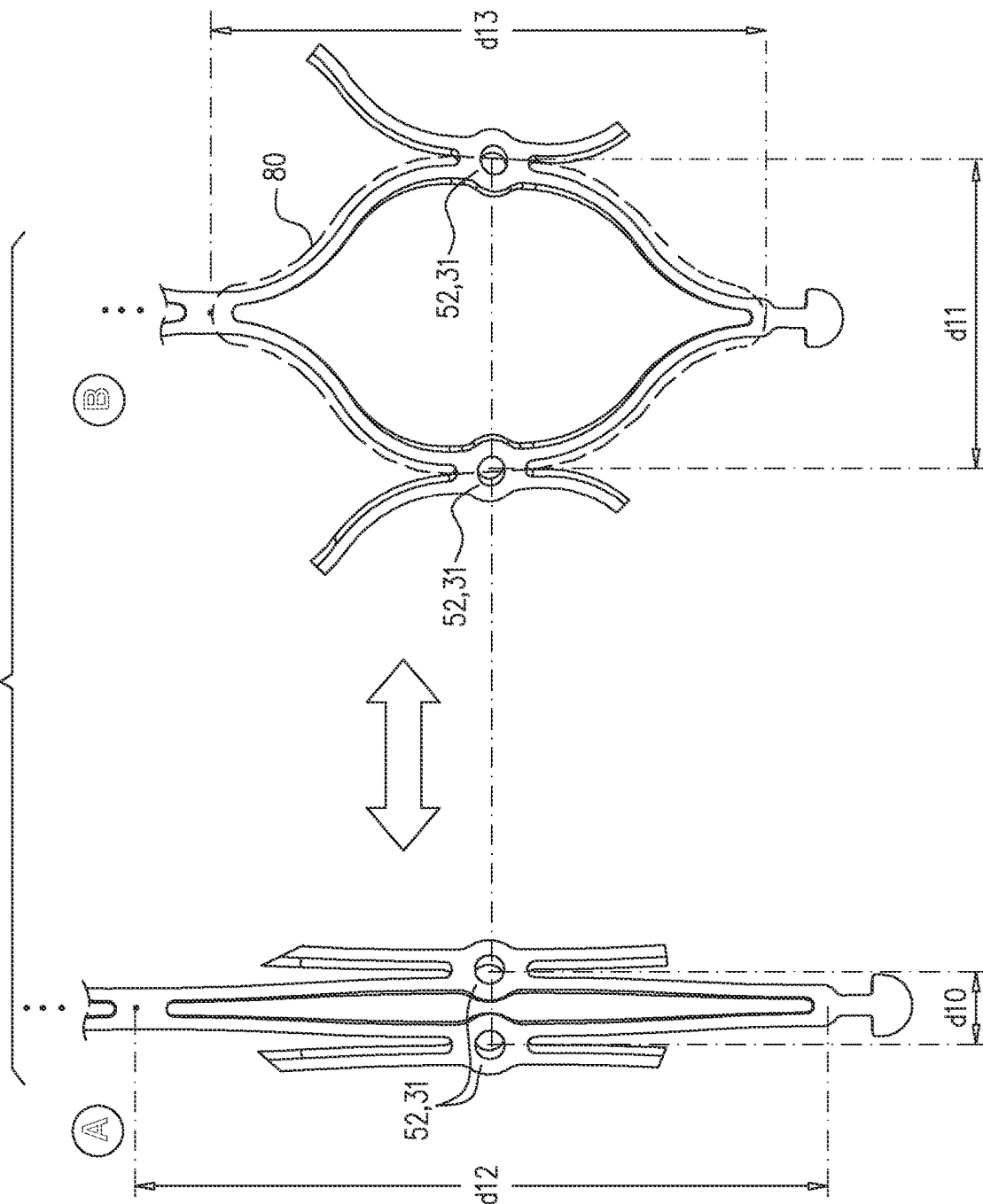
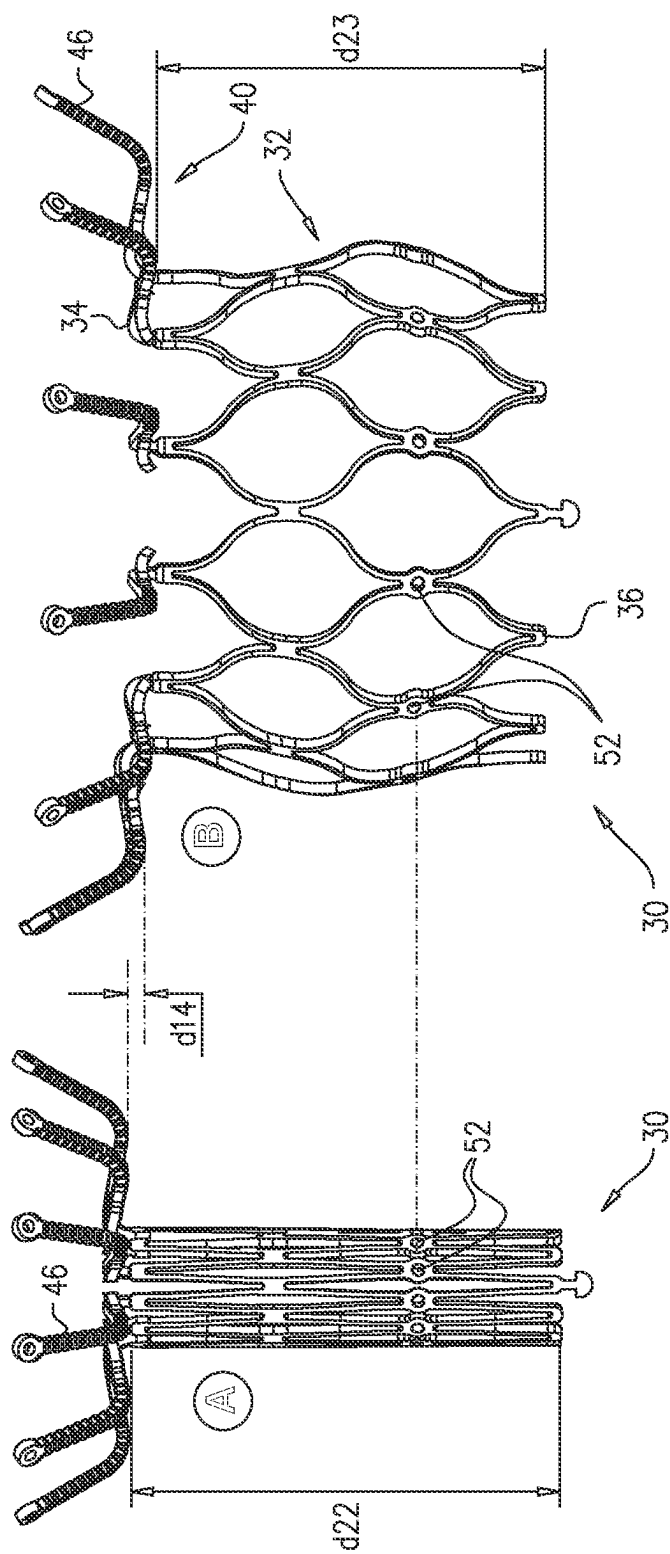
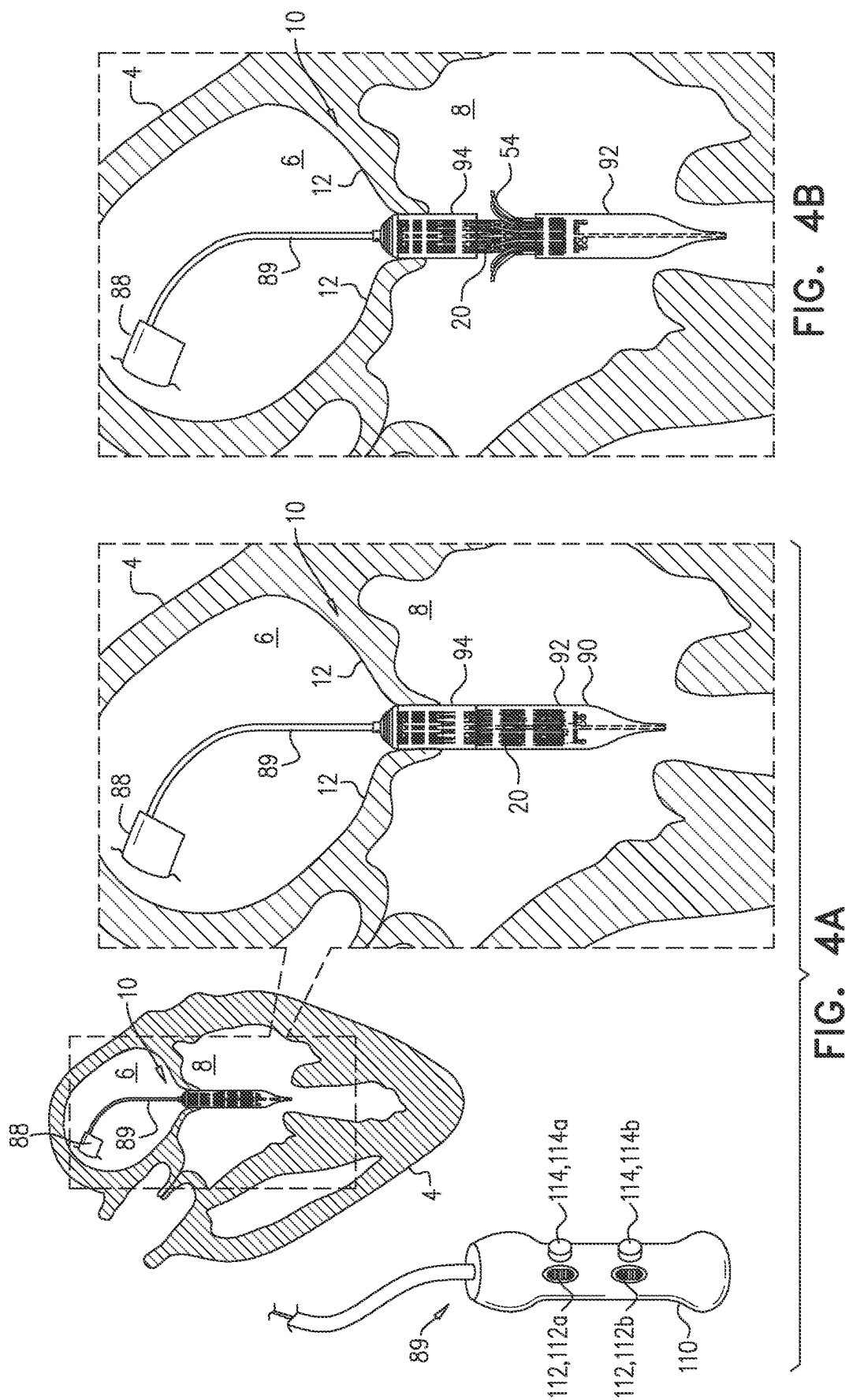


FIG. 3C





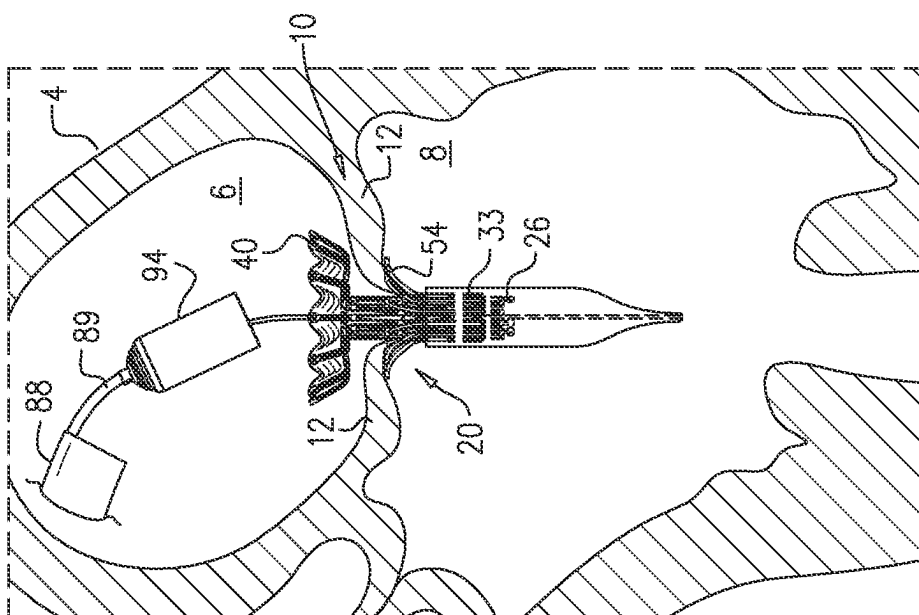


FIG. 4D

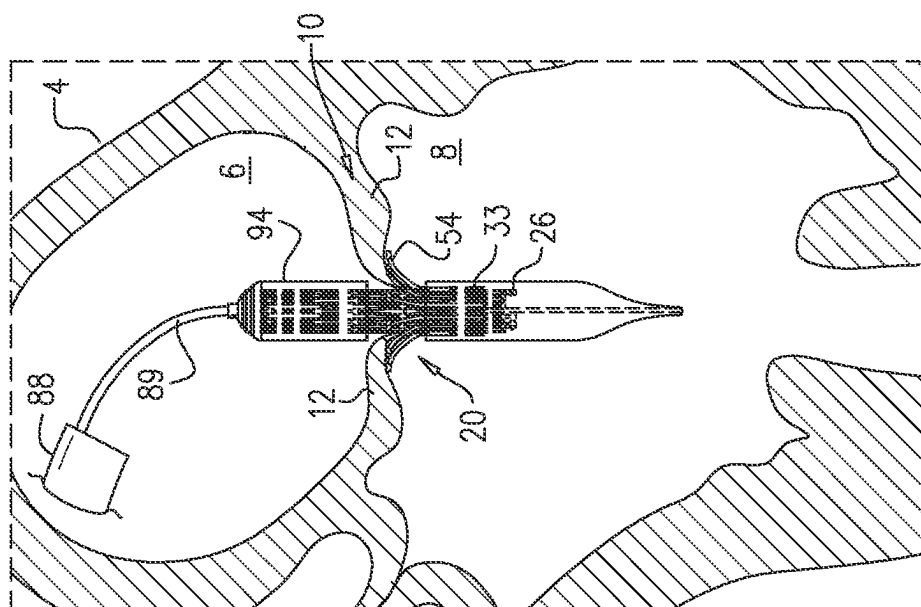
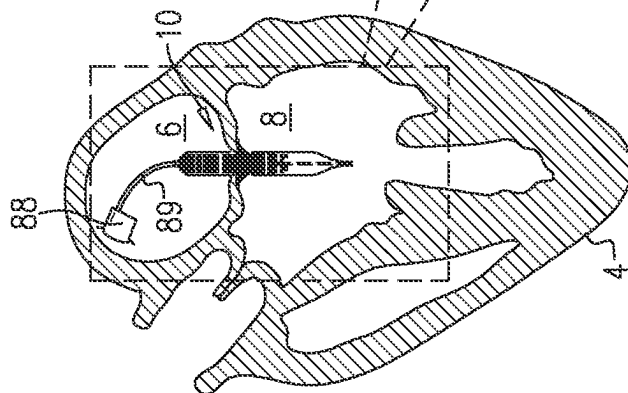


FIG. 4C



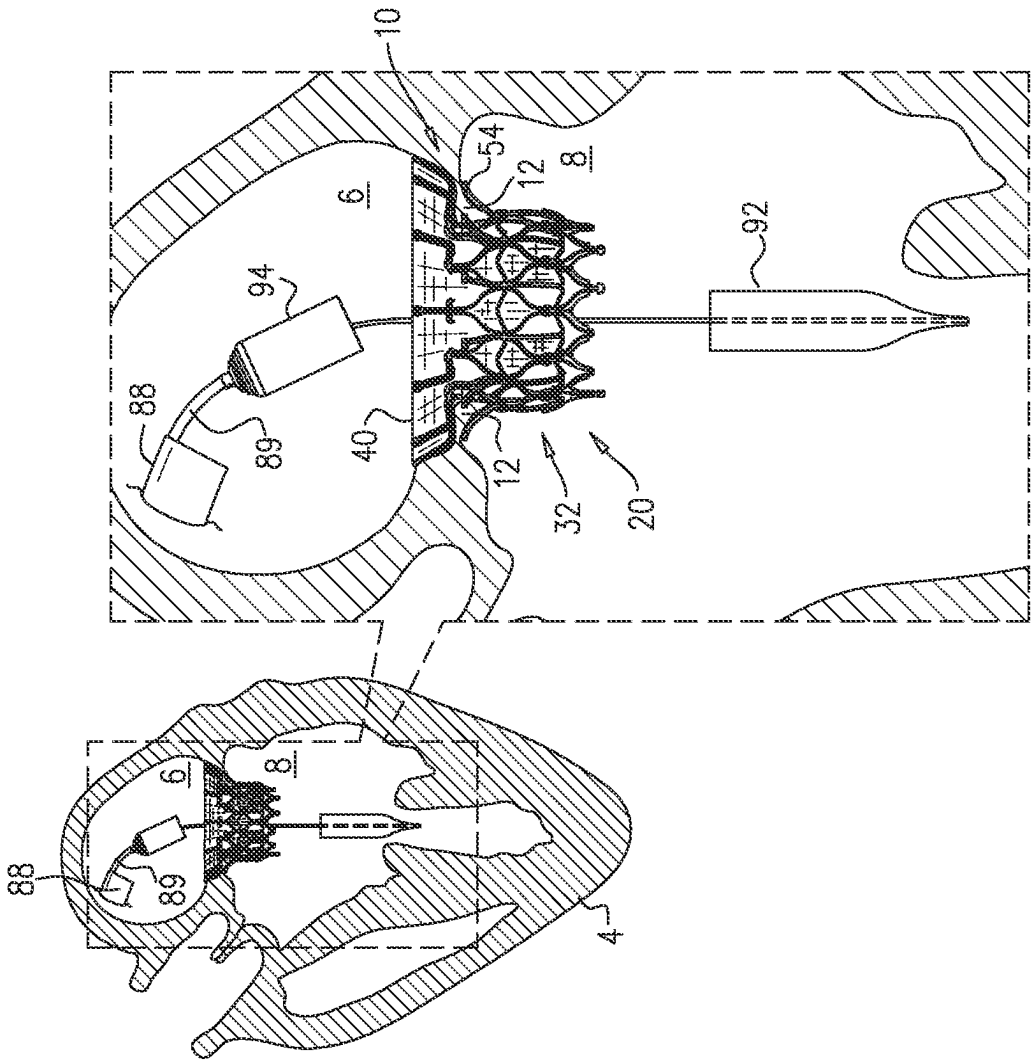


FIG. 4E

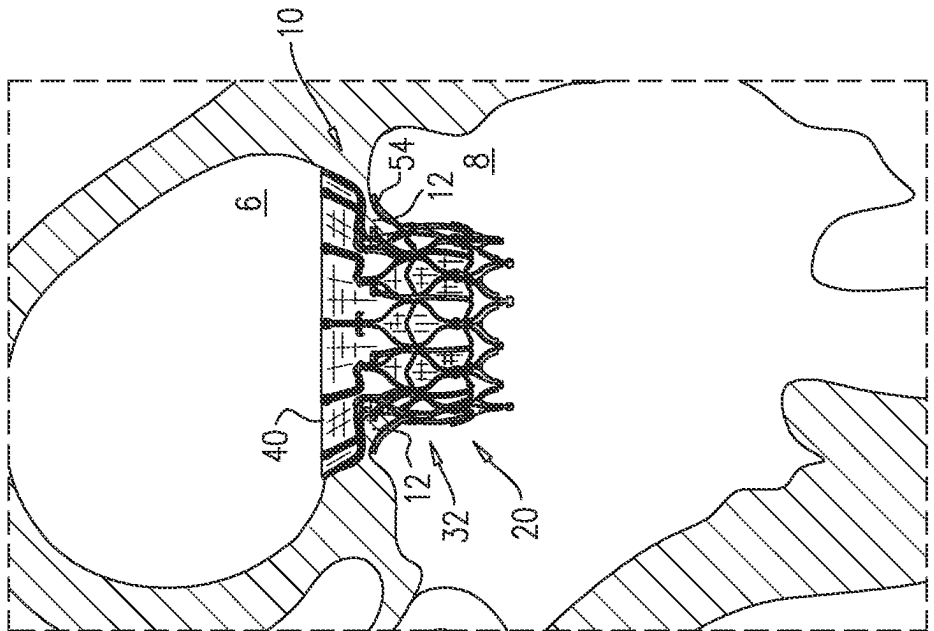


FIG. 4F

FIG. 5

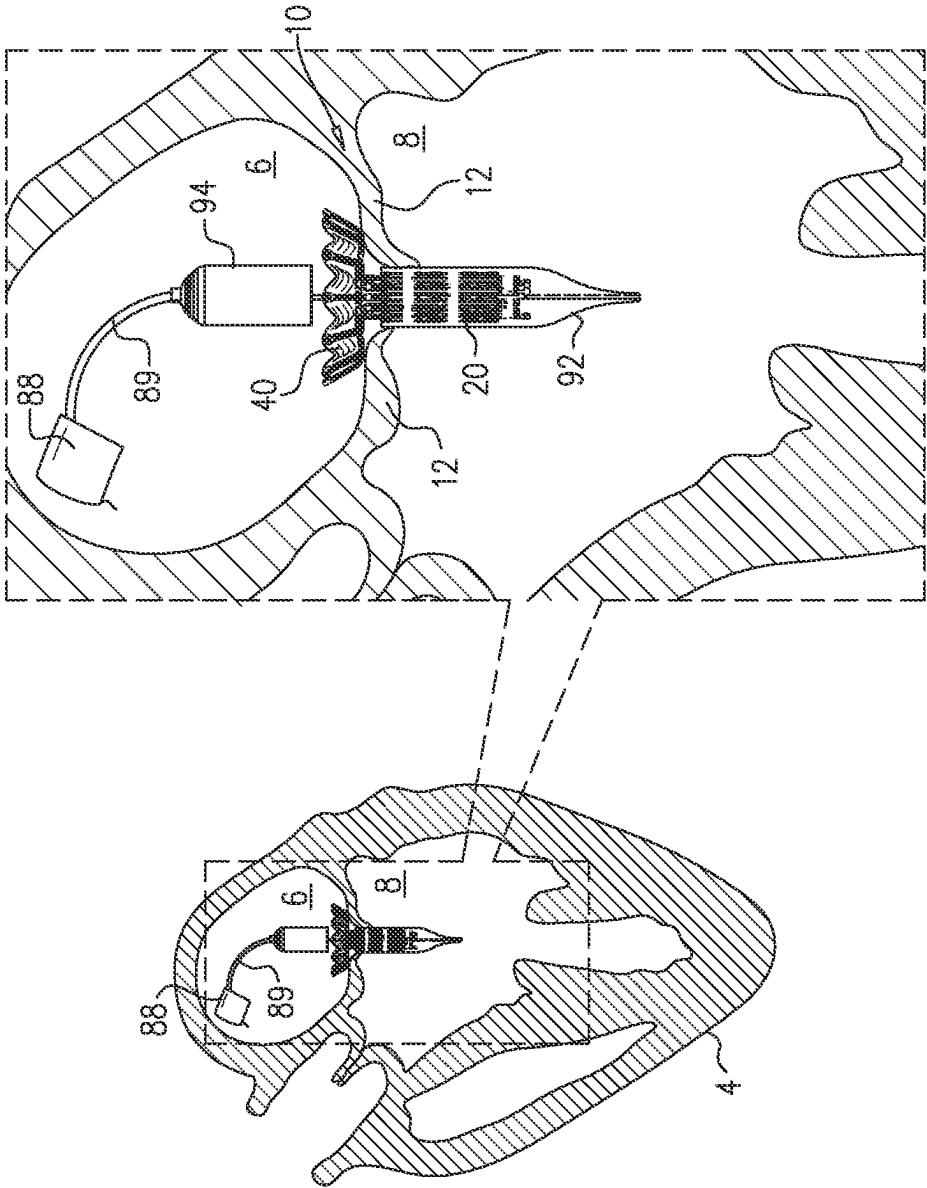


FIG. 6

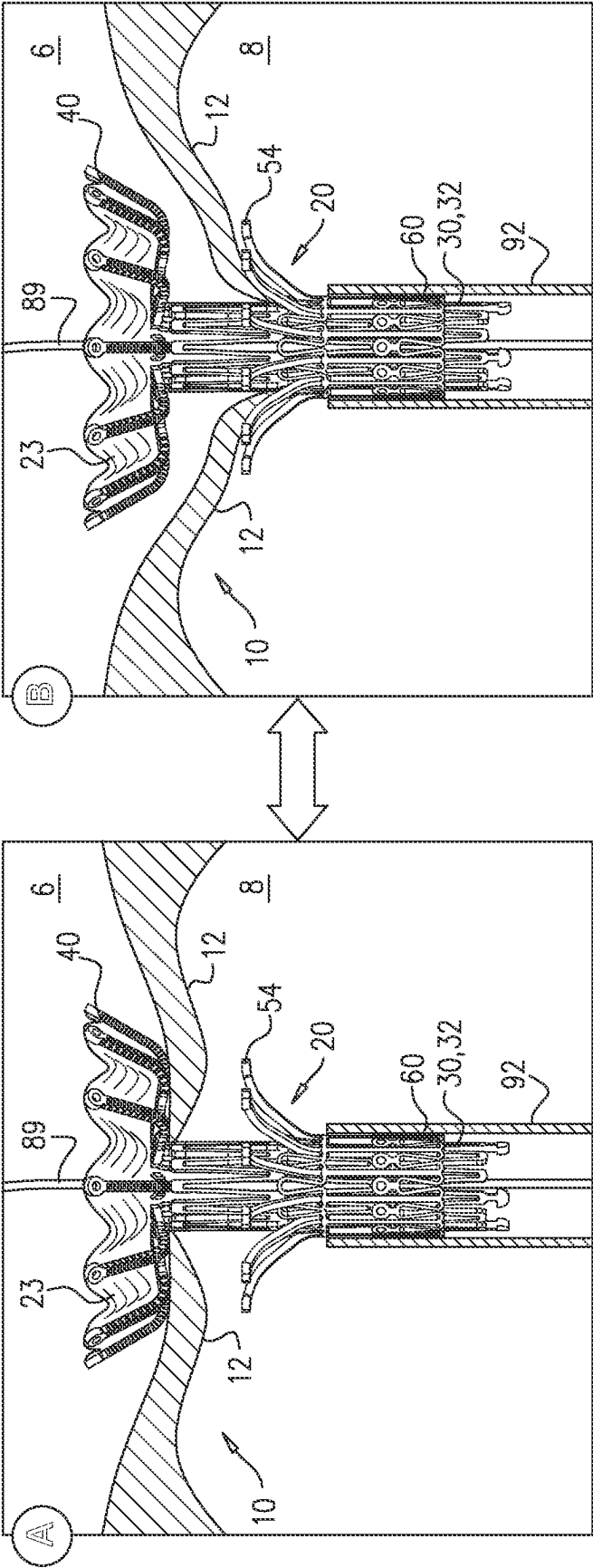


FIG. 7A

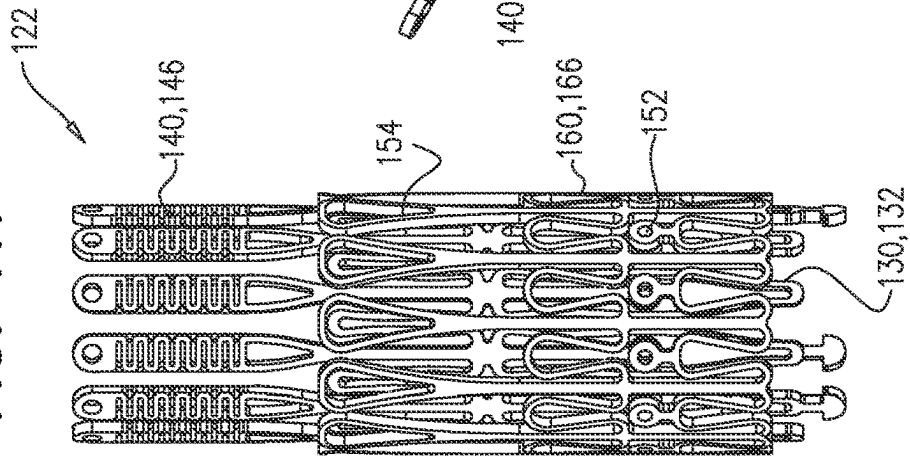


FIG. 7B

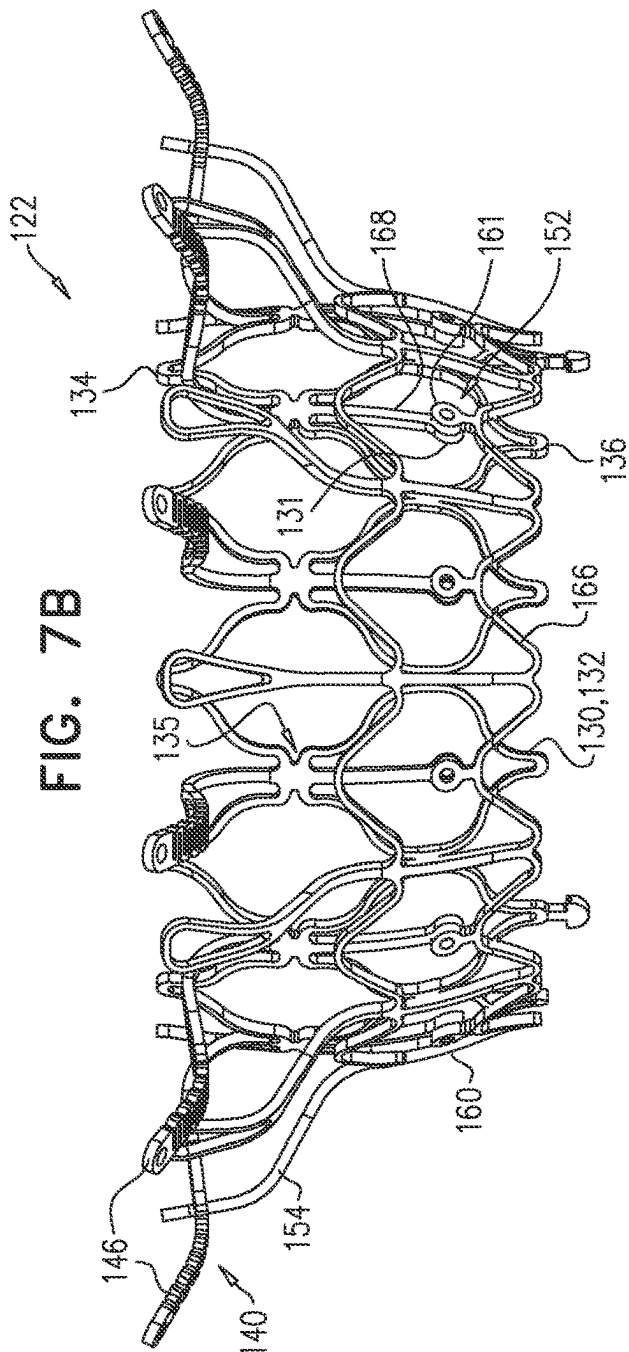


FIG. 8A

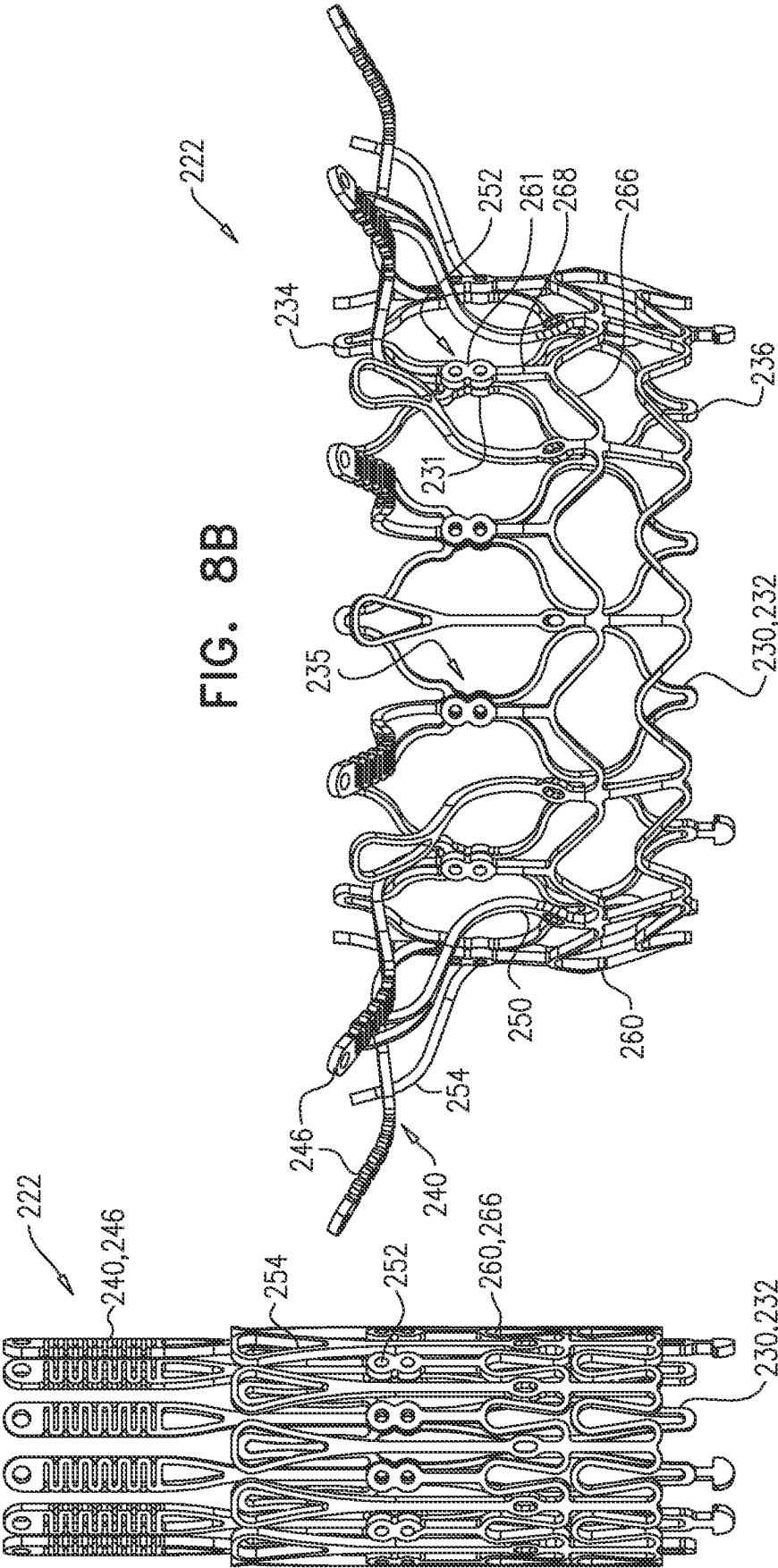
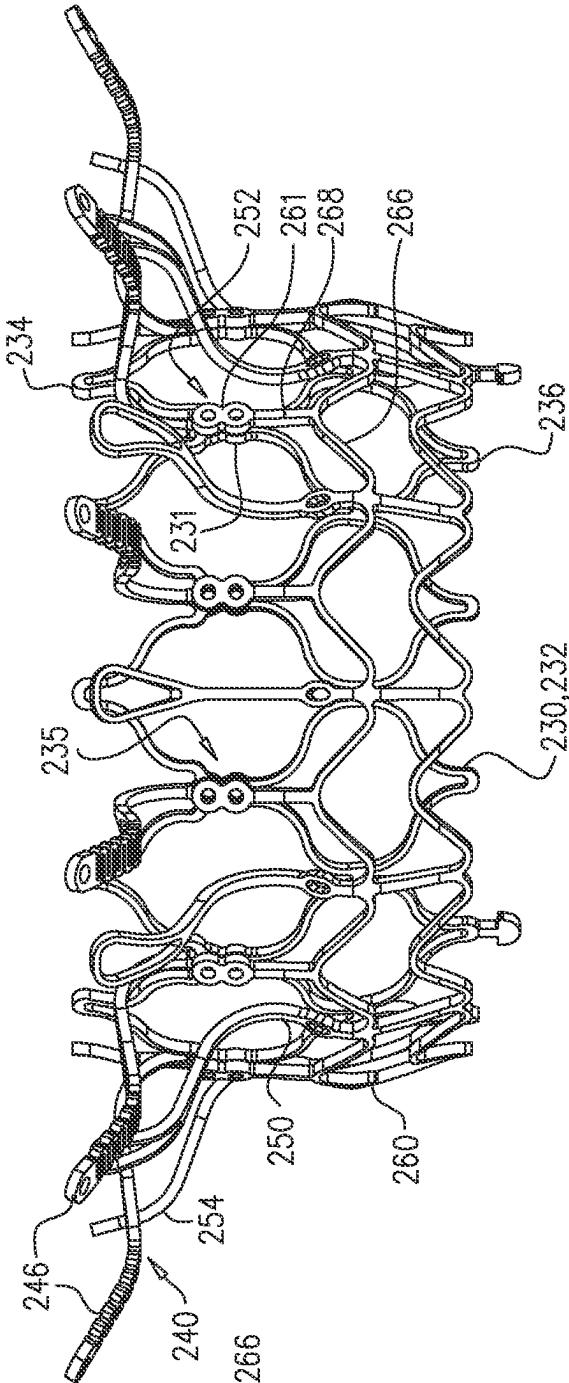


FIG. 8B



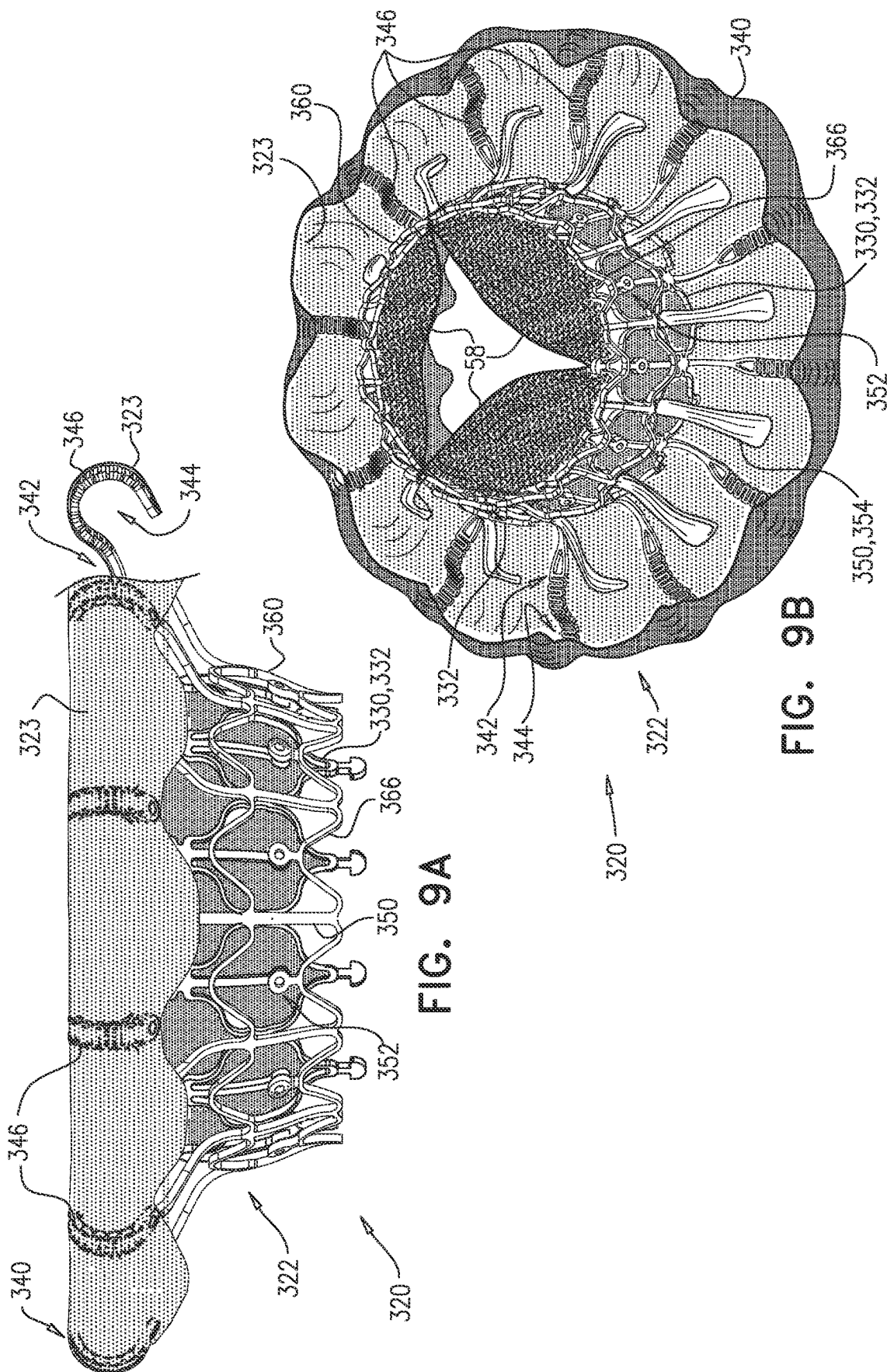
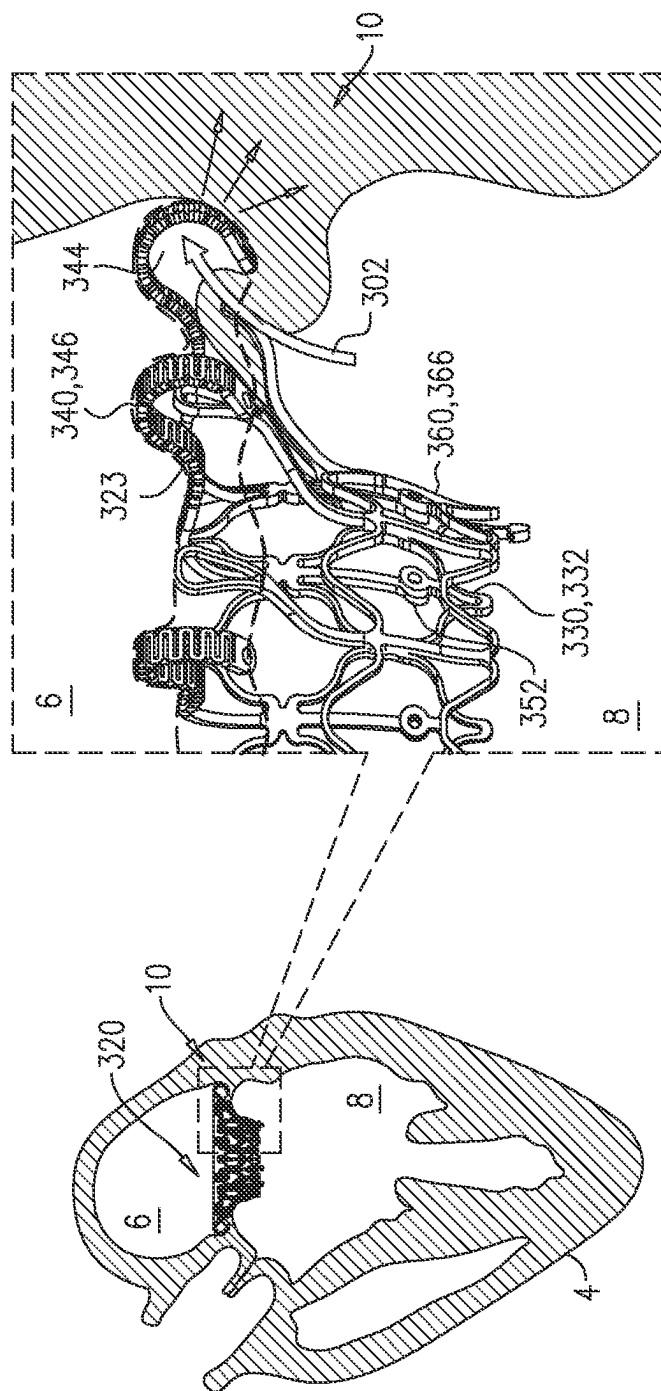


FIG. 9C



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PROSTHETIC VALVE WITH ARMS AND FLANGES**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is a Continuation application of U.S. patent application Ser. No. 17/141,853, filed Jan. 5, 2021, entitled, "Techniques for deployment of a prosthetic valve," which issued as U.S. Pat. No. 11,801,135 and which is a Continuation application of U.S. patent application Ser. No. 15/668,659, filed Aug. 3, 2017, entitled, "Techniques for deployment of a prosthetic valve," which issued as U.S. Pat. No. 10,918,481 and which:

- (1) is a Continuation-In-Part of PCT application IL2016/050125, filed Feb. 3, 2016, entitled "Prosthetic valve with axially-sliding frames," which published as WO 2016/125160, and which claims priority from U.S. Provisional Patent Application 62/112,343 to Hariton et al., filed Feb. 5, 2015, and entitled "Prosthetic valve with axially-sliding frames; and
- (2) is a Continuation-In-Part of U.S. patent application Ser. No. 15/541,783, entitled "Prosthetic valve with axially-sliding frames," (now U.S. Pat. No. 9,974,651) which is the US National Phase of PCT application IL2016/050125, filed Feb. 3, 2016, entitled "Prosthetic valve with axially-sliding frames," which published as WO 2016/125160, and which claims priority from U.S. Provisional Patent Application 62/112,343 to Hariton et al., filed Feb. 5, 2015, and entitled "Prosthetic valve with axially-sliding frames."

All of the above applications are incorporated herein by reference.

FIELD OF THE INVENTION

Some applications of the present invention relate in general to valve replacement. More specifically, some applications of the present invention relate to prosthetic valves for replacement of a cardiac valve.

BACKGROUND

Ischemic heart disease causes regurgitation of a heart valve by the combination of ischemic dysfunction of the papillary muscles, and the dilatation of the ventricle that is present in ischemic heart disease, with the subsequent displacement of the papillary muscles and the dilatation of the valve annulus.

Dilation of the annulus of the valve prevents the valve leaflets from fully coapting when the valve is closed. Regurgitation of blood from the ventricle into the atrium results in increased total stroke volume and decreased cardiac output, and ultimate weakening of the ventricle secondary to a volume overload and a pressure overload of the atrium.

SUMMARY OF THE INVENTION

For some applications, an implant is provided having a tubular portion, an upstream support portion and one or more flanges. The implant is percutaneously deliverable to a native heart valve in a compressed state, and is expandable at the native valve. The implant and its delivery system facilitate causing the upstream support portion and the flanges to protrude radially outward from the tubular portion without expanding the tubular portion. Expansion of the tubular portion brings the upstream support portion and the

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flanges closer together, for securing the implant at the native valve by sandwiching tissue of the native valve between the upstream support portion and the flanges.

For some applications, a delivery tool is provided, including a delivery capsule, disposed at a distal portion of the tool. The delivery capsule includes a proximal capsule-portion dimensioned to house a first part of an implant, and a distal capsule-portion dimensioned to house a second part of the implant; and an extracorporeal control portion, disposed at a proximal portion of the tool, the control portion including one or more controllers and one or more locks. The controllers are operatively coupled to the proximal and distal capsule-portions such that the proximal and distal capsule-portions are movable with respect to the implant via actuation of the controllers. The tool has sequential states in which movement of the proximal and distal capsule-portions is variously inhibited or facilitated. For some applications, in response to a movement of a capsule-portion that is facilitated in a given state, the tool unlocks the subsequent state.

There is therefore provided, in accordance with an application of the present invention, apparatus including an implant and a delivery tool, the tool including:

a delivery capsule, disposed at a distal portion of the tool, the delivery capsule including a proximal capsule-portion dimensioned to house a first part of the implant, and a distal capsule-portion dimensioned to house a second part of the implant; and

an extracorporeal control portion, disposed at a proximal portion of the tool, the control portion including one or more controllers and one or more locks, the controllers being operatively coupled to the proximal and distal capsule-portions such that the proximal and distal capsule-portions are movable with respect to the implant via actuation of the controllers, and the controllers and the locks are mechanically cooperative such that: in a state A of the tool:

a first distal movement of the distal capsule-portion up to a first distance distally with respect to the implant is facilitated,

a second distal movement of the distal capsule-portion further distally with respect to the implant is inhibited, and

a proximal movement of the proximal capsule-portion proximally with respect to the implant is facilitated; and

in a state B of the tool:

the second distal movement of the distal capsule-portion is facilitated.

In an application, in the state A, a lock of the one or more locks is locked, and unlocking of the lock transitions the tool into the state B.

In an application, the lock is not unlockable until the distal capsule-portion has been moved the first distance distally with respect to the implant.

In an application, the lock is not unlockable until the proximal capsule-portion has been moved proximally with respect to the implant.

In an application:

the one or more controllers include a first controller and a second controller, and

in the state A, actuation of the first controller causes the first distal movement of the distal capsule-portion, and actuation of the second controller causes the proximal movement of the proximal capsule-portion.

In an application, in the state B, further actuation of the first controller causes the second distal movement of the distal capsule-portion.

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In an application, the one or more controllers further include a third controller, and, in the state B, actuation of the third controller causes the second distal movement of the distal capsule-portion.

In an application, the implant is for use at a native mitral valve of a subject, and:

the first part of the implant includes an upstream support portion, configured to be placed against an upstream surface of the native mitral valve,

the second part of the implant includes a downstream end of the implant, and a plurality of flanges (i) disposed longitudinally between the upstream support portion and the downstream end, and (ii) configured to engage leaflets of the native mitral valve; and

the implant is housed by the delivery capsule such that the first distal movement of the distal capsule-portion unhouses, from the distal capsule-portion, the plurality of flanges, but not the downstream end of the implant.

In an application, the implant is housed by the delivery capsule such that the second distal movement of the distal capsule-portion unhouses, from the distal capsule-portion, the downstream end of the implant.

In an application, the implant is housed by the delivery capsule such that the proximal movement of the proximal capsule-portion unhouses, from the proximal capsule-portion, the upstream support portion.

In an application, the implant includes:

a valve frame that includes:

a tubular portion shaped to define a lumen there-through, and

the upstream support portion, extending from an upstream end of the tubular portion;

an outer frame that circumscribes the tubular portion, and includes the plurality of flanges; and

a plurality of prosthetic leaflets, coupled to the tubular portion within the lumen.

In an application, the second part of the implant includes at least part of the tubular portion.

There is further provided, in accordance with an application of the present invention, apparatus including an implant and a delivery tool, the tool including:

a delivery capsule, disposed at a distal portion of the tool, the delivery capsule including a proximal capsule-portion dimensioned to house a first part of the implant, and a distal capsule-portion dimensioned to house a second part of the implant; and

an extracorporeal control portion, disposed at a proximal portion of the tool, the control portion including one or more controllers and one or more locks, the controllers being operatively coupled to the proximal and distal capsule-portions such that the proximal and distal capsule-portions are movable with respect to the implant via actuation of the controllers, and the controllers and the locks are mechanically cooperative such that:

in a first state of the tool:

a first distal movement of the distal capsule-portion up to a first distance distally with respect to the implant is facilitated,

a second distal movement of the distal capsule-portion further distally with respect to the implant is inhibited,

a proximal movement of the proximal capsule-portion proximally with respect to the implant is inhibited, and

responsively to movement of the distal capsule-portion distally, the tool automatically unlocks a second state of the tool;

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in the second state of the tool:

the proximal movement of the proximal capsule-portion is facilitated,

the second distal movement of the distal capsule-portion remains inhibited, and

responsively to movement of the proximal capsule-portion proximally, the tool automatically unlocks a third state of the tool; and in the third state of the tool:

the second distal movement of the distal capsule-portion is facilitated.

In an application, the tool automatically unlocks the second state responsively to the distal capsule-portion having moved the first distance.

In an application:

the one or more locks include a first lock and a second lock,

the one or more controllers include a first controller and a second controller,

in the first state of the tool, actuation of the first controller causes the first distal movement of the distal capsule-portion,

the tool automatically unlocks the second state by automatically making the first lock unlockable,

in the second state of the tool, actuation of the second controller causes the proximal movement of the proximal capsule-portion, and

the tool automatically unlocks the third state by automatically making the second lock unlockable.

In an application, in the third state of the tool, further actuation of the first controller causes the second distal movement of the distal capsule-portion.

In an application, the one or more controllers further include a third controller, and, in the third state of the tool, actuation of the third controller causes the second distal movement of the distal capsule-portion.

In an application, the implant is for use at a native mitral valve of a subject, and:

the first part of the implant includes an upstream support portion, configured to be placed against an upstream surface of the native mitral valve,

the second part of the implant includes a downstream end of the implant, and a plurality of flanges (i) disposed longitudinally between the upstream support portion and the downstream end, and (ii) configured to engage leaflets of the native mitral valve; and

the implant is housed by the delivery capsule such that the first distal movement of the distal capsule-portion unhouses, from the distal capsule-portion, the plurality of flanges, but not the downstream end of the implant.

In an application, the implant is housed by the delivery capsule such that the second distal movement of the distal capsule-portion unhouses, from the distal capsule-portion, the downstream end of the implant.

In an application, the implant is housed by the delivery capsule such that the proximal movement of the proximal capsule-portion unhouses, from the proximal capsule-portion, the upstream support portion.

In an application, the implant includes:

a valve frame that includes:

a tubular portion shaped to define a lumen there-through, and

the upstream support portion, extending from an upstream end of the tubular portion;

an outer frame that circumscribes the tubular portion, and includes the plurality of flanges; and

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a plurality of prosthetic leaflets, coupled to the tubular portion within the lumen.

In an application, the second part of the implant includes at least part of the tubular portion.

There is further provided, in accordance with an application of the present invention, a method, including:

percutaneously advancing an implant to a native heart valve of a subject, the implant housed in a compressed state within a delivery tool;

unhousing a longitudinally-intermediate portion of the implant, the longitudinally-intermediate portion being longitudinally between a longitudinally-proximal portion of the implant and a longitudinally distal portion of the implant;

subsequently, unhousing the longitudinally-proximal portion of the implant; and

subsequently, unhousing the longitudinally-distal portion of the implant.

In an application:

unhousing the longitudinally-intermediate portion of the implant includes unhousing flanges of the implant such that the flanges automatically expand;

unhousing the longitudinally-proximal portion of the implant includes unhousing an upstream support portion of the implant such that the upstream support portion automatically expands; and

unhousing the longitudinally-distal portion of the implant includes unhousing the longitudinally-distal portion of the implant such that the longitudinally-distal portion of the implant automatically expands.

In an application, the longitudinally-distal portion includes a tubular portion within which a valve member is disposed, and unhousing the longitudinally-distal portion of the implant includes unhousing the longitudinally-distal portion of the implant such that the tubular portion expands to form a lumen and the valve member regulates one-way flow of blood through the lumen.

In an application, unhousing the longitudinally-distal portion of the implant includes facilitating expansion of the implant such that the tubular portion expands and the flanges become longitudinally closer to the upstream support portion.

There is further provided, in accordance with an application of the present invention, apparatus for use with a native valve that is disposed between an atrium and a ventricle of a heart of a subject, the apparatus including:

a valve frame, including:

a tubular portion that circumscribes a longitudinal axis of the valve frame so as to define a lumen along the axis, the tubular portion defining a plurality of valve-frame coupling elements disposed circumferentially around the longitudinal axis; and

a plurality of arms, extending radially outward from an upstream portion of the tubular portion to define an arm span, and configured to engage tissue in the atrium;

a plurality of prosthetic leaflets, coupled to the frame, disposed within the lumen, and arranged to provide unidirectional flow of blood from an upstream end of the lumen to a downstream end of the lumen; and

an outer frame:

coupled to the valve frame, and

including a ring that circumscribes the tubular portion, and a plurality of flanges that extend radially outward from the ring to define a flange span, and are configured to engage tissue of the ventricle,

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and:

each arm of the plurality of arms defines a rigid portion and a flexible portion that is disposed radially outward from the rigid portion, and is more flexible than the rigid portion, and the majority of the flexible portion is disposed further radially outward than the flange span.

In an application, the implant is configured such that the rigid portion provides resistance against axial movement of the implant, while the flexible portion facilitates conformation of the arm to anatomy of the atrium and/or lateral movement of the implant within the native valve.

In an application, each arm of the plurality of arms defines a serpentine shape in the flexible portion.

In an application, each arm of the plurality of arms defines a plurality of holes in the flexible portion.

In an application, less than 90 percent of the flexible portion is disposed further radially outward than the flange span.

In an application, more than 60 percent of the flexible portion is disposed further radially outward than the flange span.

In an application, for each arm of the plurality of arms, the flexible portion covers an overall surface area including (i) the area occupied by material from which the arm is formed, and (ii) spaces, and the material from which the arm is formed occupies less than 80 percent of the overall surface area.

In an application, the material from which the arm is formed occupies more than 20 percent of the overall surface area.

In an application, the material from which the arm is formed occupies less than 70 percent of the overall surface area.

There is further provided, in accordance with an application of the present invention, apparatus for use with a native valve that is disposed between an atrium and a ventricle of a heart of a subject, the apparatus including:

a valve frame, including a tubular portion that circumscribes a longitudinal axis of the valve frame so as to define a lumen along the axis, the tubular portion defining a plurality of valve-frame coupling elements disposed circumferentially around the longitudinal axis;

a plurality of prosthetic leaflets, coupled to the frame, disposed within the lumen, and arranged to provide unidirectional flow of blood from an upstream end of the lumen to a downstream end of the lumen;

an outer frame:

including a ring defined by a pattern of alternating peaks and troughs, the peaks being longitudinally closer to the upstream end than to the downstream end, and the troughs being longitudinally closer to the downstream end than to the upstream end, and the pattern of the ring having an amplitude longitudinally between the peaks and the troughs,

including a plurality of legs, each of the legs coupled to the ring at a respective trough, and

shaped to define a plurality of outer-frame coupling elements, each of the outer-frame coupling elements (i) coupled to the ring at a respective peak, and (ii) fixed with respect to a respective valve-frame coupling element,

and:

the tubular portion has (i) a compressed state in which the tubular portion has a compressed diameter, and (ii) an

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expanded state in which the tubular portion has an expanded diameter that is greater than the compressed diameter, and

the fixation of the outer-frame coupling elements to the valve-frame coupling elements is such that compression of the tubular portion from the expanded state toward the compressed state such that the valve-frame coupling elements pull the outer-frame coupling elements radially inward: (i) reduces a circumferential distance between each of the outer-frame coupling elements and its adjacent outer-frame coupling elements, and (ii) increases the amplitude of the pattern of the ring.

In an application, the ring circumscribes the tubular portion.

In an application, the valve-frame coupling elements are disposed circumferentially around the longitudinal axis between the upstream end and the downstream end but not at the upstream end nor at the downstream end.

In an application, the upstream support portion includes one or more fabric pockets disposed circumferentially, each pocket of the one or more pockets having an opening that faces a downstream direction.

In an application, the outer frame is coupled to the valve frame only via the fixation of the outer-frame coupling elements to the respective valve-frame coupling elements.

In an application, the apparatus further includes an upstream support portion that includes a plurality of arms that extend radially from the tubular portion, and:

the upstream support portion has (i) a constrained-arm state, and (ii) a released-arm state in which the arms extend radially outward from the tubular portion, each leg has a tissue-engaging flange that has (i) a constrained-flange state, and (ii) a released-flange state in which the flange extends radially outward from the tubular portion, and

the apparatus has an intermediate state in which (i) the tubular portion is in its compressed state, (ii) the upstream support portion is in its released-arm state, and (iii) the legs are in their released-flange state.

In an application:

the apparatus includes an implant that includes the valve frame, the leaflets, and the outer frame, and

the apparatus further includes a tool:

including a delivery capsule dimensioned (i) to house and retain the implant in a compressed state of the implant in which (a) the tubular portion is in its compressed state, (b) the upstream support portion is in its constrained-arm state, and (c) the legs are in their constrained-flange state, and (ii) to be advanced percutaneously to the heart of the subject while the implant is housed and in its compressed state, and operable from outside the subject to:

transition the implant from its compressed state into the intermediate state while retaining the tubular portion in its compressed state, and subsequently, expand the tubular portion toward its expanded state.

In an application, the tool is operable from outside the subject to transition the implant from its compressed state into the intermediate state by (i) releasing the legs into their released-flange state, while retaining the tubular portion in its compressed state, and (ii) subsequently, releasing the upstream support portion into its released-arm state, while retaining the tubular portion in its compressed state.

In an application, the tool is operable from outside the subject to transition the implant from its compressed state

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into the intermediate state by (i) releasing the upstream support portion into its released-arm state, while retaining the tubular portion in its compressed state, and (ii) subsequently, releasing the legs into their released-flange state, while retaining the tubular portion in its compressed state.

In an application, the fixation of the outer-frame coupling elements to the valve-frame coupling elements is such that, when the apparatus is in its intermediate state, expansion of the tubular portion from its compressed state toward its expanded state moves the flanges longitudinally away from the valve-frame coupling elements.

In an application, the fixation of the outer-frame coupling elements to the valve-frame coupling elements is such that, when the apparatus is in its intermediate state, expansion of the tubular portion from its compressed state toward its expanded state reduces the amplitude of the pattern of the ring and passes the flanges between the arms.

In an application, the upstream support portion further includes a covering that covers the arms to form an annular shape in the released-arm state, and, when the apparatus is in its intermediate state, expansion of the tubular portion from its compressed state toward its expanded state presses the flanges onto the covering.

In an application, in the compressed state of the tubular portion, a downstream end of each leg is longitudinally closer than the valve-frame coupling elements to the downstream end, and the flange of each leg is disposed longitudinally closer than the valve-frame coupling elements to the upstream end.

In an application, in the expanded state of the tubular portion, the downstream end of each leg is longitudinally closer than the valve-frame coupling elements to the downstream end, and the flange of each leg is disposed longitudinally closer than the valve-frame coupling elements to the upstream end.

There is further provided, in accordance with an application of the present invention, apparatus for use with a native valve of a heart of a subject, the apparatus including an implant that includes:

a valve frame that includes a tubular portion that circumscribes a longitudinal axis of the valve frame so as to define a lumen along the axis, the tubular portion having an upstream end, a downstream end, a longitudinal length therebetween, and a diameter transverse to the longitudinal axis;

a valve member, coupled to the tubular portion, disposed within the lumen, and arranged to provide unidirectional upstream-to-downstream flow of blood through the lumen;

an upstream support portion, coupled to the tubular portion; and

an outer frame, coupled to the tubular portion, and including a tissue-engaging flange,

and:

the implant has a first state and a second state, in both the first state and the second state, (i) the upstream support portion extends radially outward from the tubular portion, and (ii) the tissue-engaging flange extends radially outward from the tubular portion, and the tubular portion, the upstream support portion, and the outer frame are arranged such that transitioning of the implant from the first state toward the second state: increases the diameter of the tubular portion by a diameter-increase amount, decreases the length of the tubular portion by a length-decrease amount, and

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moves the flange a longitudinal distance toward or toward-and-beyond the upstream support portion, the distance being greater than the length-decrease amount.

In an application, the tubular portion, the upstream support portion, and the outer frame are arranged such that the longitudinal distance is more than 20 percent greater than the length-decrease amount.

In an application, the tubular portion, the upstream support portion, and the outer frame are arranged such that the longitudinal distance is more than 30 percent greater than the length-decrease amount.

In an application, the tubular portion, the upstream support portion, and the outer frame are arranged such that the longitudinal distance is more than 40 percent greater than the length-decrease amount.

There is further provided, in accordance with an application of the present invention, apparatus for use with a native valve that is disposed between an atrium and a ventricle of a heart of a subject, the apparatus including:

a valve frame, including a tubular portion that circumscribes a longitudinal axis of the valve frame so as to define a lumen along the axis;

a plurality of prosthetic leaflets, coupled to the frame, disposed within the lumen, and arranged to provide unidirectional flow of blood from an upstream end of the lumen to a downstream end of the lumen;

an outer frame, including:

a ring defined by a pattern of alternating peaks and troughs:

the peaks being longitudinally closer than the troughs to the upstream end,

the peaks being fixed to respective sites of the tubular portion at respective coupling points disposed circumferentially around the longitudinal axis, and

the pattern of the ring having an amplitude longitudinally between the peaks and the troughs; and

a plurality of legs, each of the legs coupled to the ring at a respective trough,

and:

the tubular portion has (i) a compressed state in which the tubular portion has a compressed diameter, and (ii) an expanded state in which the tubular portion has an expanded diameter that is greater than the compressed diameter, and

the fixation of the peaks to the respective sites of the tubular portion is such that compression of the tubular portion from the expanded state toward the compressed state such that the respective sites of the tubular portion pull the peaks radially inward via radially-inward tension on the coupling points: (i) reduces a circumferential distance between each of the coupling points and its adjacent coupling points, and (ii) increases the amplitude of the pattern of the ring.

In an application, the outer frame is coupled to the valve frame only via the fixation of the peaks to the respective sites of the tubular portion at the respective coupling points.

There is further provided, in accordance with an application of the present invention, apparatus for use with a native valve that is disposed between an atrium and a ventricle of a heart of a subject, the apparatus including:

a valve frame, including a tubular portion that circumscribes a longitudinal axis of the valve frame so as to define a lumen along the axis, the valve frame defining a plurality of valve-frame coupling elements disposed circumferentially around the longitudinal axis;

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a plurality of prosthetic leaflets, coupled to the frame, disposed within the lumen, and arranged to provide unidirectional flow of blood from an upstream end of the lumen to a downstream end of the lumen;

an outer frame:

including a ring defined by a pattern of alternating peaks and troughs, the peaks being longitudinally closer to the upstream end than to the downstream end, and the troughs being longitudinally closer to the downstream end than to the upstream end, and the pattern of the ring having an amplitude longitudinally between the peaks and the troughs,

including a plurality of legs, each of the legs coupled to the ring at a respective trough, and

shaped to define a plurality of outer-frame coupling elements, each of the outer-frame coupling elements (i) coupled to the ring at a respective peak, and (ii) fixed with respect to a respective valve-frame coupling element,

and:

the tubular portion has (i) a compressed state in which the tubular portion has a compressed diameter, and (ii) an expanded state in which the tubular portion has an expanded diameter that is greater than the compressed diameter, and

the fixation of the outer-frame coupling elements with respect to the valve-frame coupling elements is such that compression of the tubular portion from the expanded state toward the compressed state (i) pulls the outer-frame coupling elements radially inward via radially-inward pulling of the valve-frame coupling elements on the outer-frame coupling elements, (ii) reduces a circumferential distance between each of the outer-frame coupling elements and its adjacent outer-frame coupling elements, and (iii) increases the amplitude of the pattern of the ring, without increasing a radial gap between the valve frame and the ring by more than 1.5 mm.

In an application, the outer frame is coupled to the valve frame only via the fixation of the outer-frame coupling elements to the respective valve-frame coupling elements.

There is further provided, in accordance with an application of the present invention, apparatus for use with a native valve that is disposed between an atrium and a ventricle of a heart of a subject, the apparatus including:

a valve frame, including a tubular portion that circumscribes a longitudinal axis of the valve frame so as to define a lumen along the axis;

a plurality of prosthetic leaflets, coupled to the frame, disposed within the lumen, and arranged to provide unidirectional flow of blood from an upstream end of the lumen to a downstream end of the lumen;

an outer frame, including:

a ring defined by a pattern of alternating peaks and troughs:

the peaks being longitudinally closer than the troughs to the upstream end,

the peaks being fixed to respective sites of the tubular portion at respective coupling points disposed circumferentially around the longitudinal axis, and

the pattern of the ring having an amplitude longitudinally between the peaks and the troughs; and

a plurality of legs, each of the legs coupled to the ring at a respective trough,

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and:

the tubular portion has (i) a compressed state in which the tubular portion has a compressed diameter, and (ii) an expanded state in which the tubular portion has an expanded diameter that is greater than the compressed diameter, and

the fixation of the peaks to the respective sites of the tubular portion is such that compression of the tubular portion from the expanded state toward the compressed state (i) pulls the peaks radially inward via radially-inward pulling of the respective sites of the tubular portion on the peaks, (ii) reduces a circumferential distance between each of the coupling points and its adjacent coupling points, and (iii) increases the amplitude of the pattern of the ring, without increasing a radial gap between the valve frame and the ring by more than 1.5 mm.

In an application, the outer frame is coupled to the valve frame only via the fixation of the peaks to the respective sites of the tubular portion at the respective coupling points.

There is further provided, in accordance with an application of the present invention, apparatus for use with a native valve disposed between an atrium and a ventricle of a heart of a subject, the apparatus including:

a valve frame, including a tubular portion that circumscribes a longitudinal axis of the valve frame so as to define a lumen along the axis, the tubular portion having an upstream end, a downstream end, and defining a plurality of valve-frame coupling elements disposed circumferentially around the longitudinal axis between the upstream end and the downstream end but not at the upstream end nor at the downstream end;

a plurality of prosthetic leaflets, disposed within the lumen, and arranged to provide unidirectional flow of blood through the lumen;

an outer frame:

including a ring defined by a pattern of alternating peaks and troughs, the peaks being longitudinally closer to the upstream end than to the downstream end, and the troughs being longitudinally closer to the downstream end than to the upstream end, including a plurality of legs, each of the legs coupled to the ring at a respective trough, and shaped to define a plurality of outer-frame coupling elements, each of the outer-frame coupling elements (i) coupled to the ring at a respective peak, and (ii) fixed with respect to a respective valve-frame coupling element at a respective coupling point,

and:

the tubular portion has (i) a compressed state in which the tubular portion has a compressed diameter, and (ii) an expanded state in which the tubular portion has an expanded diameter that is greater than the compressed diameter, and

expansion of the tubular portion from the compressed state toward the expanded state (i) increases a circumferential distance between each of the outer-frame coupling elements and its adjacent outer-frame coupling elements, and (ii) moves the plurality of legs in a longitudinally upstream direction with respect to the tubular portion.

In an application, the outer frame is coupled to the valve frame only via the fixation of the outer-frame coupling elements to the respective valve-frame coupling elements.

There is further provided, in accordance with an application of the present invention, apparatus for use with a native

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valve disposed between an atrium and a ventricle of a heart of a subject, the apparatus including:

a valve frame, including a tubular portion that circumscribes a longitudinal axis of the valve frame so as to define a lumen along the axis, the tubular portion having an upstream end and a downstream end;

a plurality of prosthetic leaflets, disposed within the lumen, and arranged to provide unidirectional flow of blood through the lumen;

an outer frame, including:

a ring defined by a pattern of alternating peaks and troughs:

the peaks being longitudinally closer than the troughs to the upstream end,

the peaks being fixed to respective sites of the tubular portion at respective coupling points disposed circumferentially around the longitudinal axis between the upstream end and the downstream end but not at the upstream end nor the downstream end; and

a plurality of legs, each of the legs coupled to the ring at a respective trough, and:

the tubular portion has (i) a compressed state in which the tubular portion has a compressed diameter, and (ii) an expanded state in which the tubular portion has an expanded diameter that is greater than the compressed diameter, and

expansion of the tubular portion from the compressed state toward the expanded state (i) increases a circumferential distance between each of the coupling points and its adjacent coupling points, and (ii) moves the plurality of legs in a longitudinally upstream direction with respect to the tubular portion.

In an application, the outer frame is coupled to the valve frame only via the fixation of the peaks to the respective sites of the tubular portion at the respective coupling points.

There is further provided, in accordance with an application of the present invention, apparatus for use with a native valve of a heart of a subject, the apparatus including:

a frame assembly, having an upstream end and a downstream end, and a central longitudinal axis therebetween, and including:

a valve frame, including:

a tubular portion having an upstream end and a downstream end, and shaped to define a lumen therebetween, and

an upstream support portion, extending from the upstream end of the tubular portion; and

at least one leg, coupled to the valve frame at a coupling point, and having a tissue-engaging flange; and

a valve member disposed within the lumen, and configured to facilitate one-way liquid flow through the lumen from the upstream end of the tubular portion to the downstream end of the tubular portion,

and the frame assembly:

has a compressed state, for percutaneous delivery to the heart, in which the tubular portion has a compressed diameter,

is biased to assume an expanded state in which the tubular portion has an expanded diameter that is greater than the compressed diameter, and

is configured such that increasing the diameter of the tubular portion toward the expanded diameter causes longitudinal movement:

of the upstream support portion toward the coupling point, and

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of the tissue-engaging flange away from the coupling point.

In an application:

the apparatus includes an implant that includes the frame assembly and the valve member, and

the apparatus further includes a tool:

including a delivery capsule dimensioned (i) to house and retain the implant in the compressed state, and (ii) to be advanced percutaneously to the heart of the subject while the implant is housed and in the compressed state, and

operable from outside the subject to facilitate an increase of the diameter of the tubular portion from the compressed diameter toward the expanded diameter such that the increase of the diameter actuates longitudinal movement:

of the upstream support portion toward the coupling point, and

of the tissue-engaging flange away from the coupling point.

In an application, the frame assembly is configured such that increasing the diameter of the tubular portion by expanding the frame assembly toward the expanded state causes longitudinal movement of the upstream end of the tubular portion toward the coupling point.

In an application, the coupling point is disposed closer to the downstream end of the frame assembly than are either the tissue-engaging flange or the upstream support portion.

In an application, in the expanded state of the frame assembly, the leg extends away from the central longitudinal axis.

In an application:

the expanded state of the frame assembly is a fully-expanded state of the frame assembly,

the leg is expandable into an expanded state of the leg, independently of increasing the diameter of the tubular portion, and

in the expanded state of the leg, the leg extends away from the central longitudinal axis.

In an application:

in the expanded state of the frame assembly, the leg extends away from the central longitudinal axis, and

in the compressed state of the frame assembly, the leg is generally parallel with the central longitudinal axis.

In an application, the frame assembly is configured such that the longitudinal movement of the tissue-engaging flange away from the coupling point is a translational movement of the tissue-engaging flange that does not include rotation of the tissue-engaging flange.

In an application, the frame assembly is configured such that increasing the diameter of the tubular portion by expanding the frame assembly toward the expanded state causes 1-20 mm of longitudinal movement of the tissue-engaging flange away from the coupling point.

In an application, the frame assembly is configured such that increasing the diameter of the tubular portion by expanding the frame assembly toward the expanded state causes 1-20 mm of longitudinal movement of the upstream support portion toward the coupling point.

In an application, the frame assembly is configured such that increasing the diameter of the tubular portion by expanding the frame assembly toward the expanded state reduces a distance between the upstream support portion and the tissue-engaging flange by 5-30 mm.

In an application, the frame assembly is configured such that increasing the diameter of the tubular portion by

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expanding the frame assembly toward the expanded state moves the tissue-engaging flange longitudinally past the upstream support portion.

In an application:

the tubular portion is defined by a plurality of cells of the valve frame, and

increasing the diameter of the tubular portion by expanding the frame assembly toward the expanded state:

includes (i) increasing a width, orthogonal to the longitudinal axis of the frame assembly, of each cell, and (ii) reducing a height, parallel with the longitudinal axis of the frame assembly, of each cell, and

causes longitudinal movement of the upstream support portion toward the coupling point by reducing a height, parallel with the longitudinal axis of the frame assembly, of the tubular portion, by reducing the height of each cell.

In an application, the leg is disposed on an outside of the tubular portion.

In an application:

the at least one leg includes a plurality of legs, the coupling point includes a plurality of coupling points, and

the frame assembly includes a leg frame that circumscribes the tubular portion, includes the plurality of legs, and is coupled to the valve frame at the plurality of coupling points, such that the plurality of legs is distributed circumferentially around the tubular portion.

In an application, the plurality of coupling points is disposed circumferentially around the frame assembly on a transverse plane that is orthogonal to the longitudinal axis of the frame assembly.

In an application, the plurality of legs is coupled to the valve frame via a plurality of struts, each strut:

having a first end that is coupled to a leg of the plurality of legs, and a second end that is coupled to a coupling point of the plurality of coupling points,

in the compressed state of the frame assembly, being disposed at a first angle in which the first end is disposed closer to the downstream end of the frame assembly than is the second end, and

being deflectable with respect to the coupling point of the plurality of coupling points, such that increasing the diameter of the tubular portion by expanding the frame assembly toward the expanded state causes the strut to deflect to a second angle in which the first end is disposed further from the downstream end of the frame assembly than is the first end in the compressed state of the frame assembly.

In an application, the leg frame is structured such that each leg of the plurality of legs is coupled to two struts of the plurality of struts, and two struts of the plurality of struts are coupled to each coupling point of the plurality of coupling points.

In an application, the leg is coupled to the valve frame via a strut, the strut:

having a first end that is coupled to the leg, and a second end that is coupled to the coupling point,

in the compressed state of the frame assembly, being disposed at a first angle in which the first end is disposed closer to the downstream end of the frame assembly than is the second end, and

being deflectable with respect to the coupling point, such that increasing the diameter of the tubular portion by expanding the frame assembly toward the expanded state causes the strut to deflect to a second angle in

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which the first end is disposed further from the downstream end of the frame assembly than is the first end in the compressed state of the frame assembly.

In an application, the at least one leg includes at least a first leg and a second leg.

In an application, the first leg and the second leg are both coupled to the valve frame at the coupling point.

In an application, the first leg is coupled to the coupling point via a respective first strut, and the second leg is coupled to the coupling point via a respective second strut.

In an application, the first and second legs, the first and second struts, and the coupling point are arranged such that, in the expanded state of the frame assembly:

the coupling point is disposed, circumferentially with respect to the tubular portion, between the first strut and the second strut,

the first strut is disposed, circumferentially with respect to the tubular portion, between the coupling point and the first leg, and

the second strut is disposed, circumferentially with respect to the tubular portion, between the coupling point and the second leg.

In an application, the coupling point includes at least a first coupling point and a second coupling point.

In an application, the leg is coupled to the valve frame at the first coupling point and at the second coupling point.

In an application, the leg is coupled to the first coupling point via a respective first strut, and to the second coupling point via a respective second strut.

In an application, the first and second legs, the first and second struts, and the coupling point are arranged such that, in the expanded state of the frame assembly:

the leg is disposed, circumferentially with respect to the tubular portion, between the first strut and the second strut,

the first strut is disposed, circumferentially with respect to the tubular portion, between the leg and the first coupling point, and

the second strut is disposed, circumferentially with respect to the tubular portion, between the leg and the second coupling point.

In an application, in the expanded state of the frame assembly, the upstream support portion extends radially outward from the tubular portion.

In an application: the expanded state of the frame assembly is a fully-expanded state of the frame assembly,

the upstream support portion is expandable into an expanded state of the upstream support portion, independently of increasing the diameter of the tubular portion, and

in the expanded state of the upstream support portion, the upstream support portion extends radially outward from the tubular portion.

In an application, in the compressed state of the frame assembly, the upstream support portion is generally tubular, collinear with the tubular portion, and disposed around the central longitudinal axis.

In an application, in the expanded state of the frame assembly, an inner region of the upstream support portion extends radially outward from the tubular portion at a first angle with respect to the tubular portion, and an outer region of the upstream support portion extends, from the inner region of the upstream support portion, further radially outward from the tubular portion at a second angle with respect to the tubular portion, the second angle being smaller than the first angle.

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There is further provided, in accordance with an application of the present invention, apparatus for use with a native valve of a heart of a subject, the apparatus including:

a frame assembly, having an upstream end and a downstream end, and a central longitudinal axis therebetween, and including:

a valve frame, including:

a tubular portion having an upstream end and a downstream end, and shaped to define a lumen therebetween, and

an upstream support portion, extending from the upstream end of the tubular portion; and

at least one leg, coupled to the valve frame at a coupling point, and having a tissue-engaging flange; and

a valve member disposed within the lumen, and configured to facilitate one-way liquid flow through the lumen from the upstream end of the tubular portion to the downstream end of the tubular portion,

and the frame assembly:

has a compressed state, for percutaneous delivery to the heart, in which the tubular portion has a compressed diameter,

is biased to assume an expanded state in which the tubular portion has an expanded diameter that is greater than the compressed diameter, and

is configured such that reducing the diameter of the tubular portion toward the compressed diameter causes longitudinal movement:

of the upstream support portion away from the coupling point, and

of the tissue-engaging flange toward the coupling point.

There is further provided, in accordance with an application of the present invention, apparatus for use with a native valve of a heart of a subject, the apparatus including:

a frame assembly, having an upstream end and a downstream end, and a central longitudinal axis therebetween, including:

a valve frame, including:

a tubular portion having an upstream end and a downstream end, and shaped to define a lumen therebetween, and

an upstream support portion, extending from the upstream end of the tubular portion; and

at least one leg, coupled to the valve frame at a coupling point, and having a tissue-engaging flange; and

a valve member disposed within the lumen, and configured to facilitate one-way liquid flow through the lumen from the upstream end of the tubular portion to the downstream end of the tubular portion,

and the frame assembly:

has a compressed state, for percutaneous delivery to the heart,

is intracorporeally expandable into an expanded state in which a diameter of the tubular portion is greater than in the compressed state, and

is configured such that increasing the diameter of the tubular portion by expanding the frame assembly toward the expanded state causes longitudinal movement of the tissue-engaging flange away from the coupling point.

There is further provided, in accordance with an application of the present invention, apparatus for use with a native valve of a heart of a subject, the apparatus including:

a frame assembly, having an upstream end and a downstream end, and a central longitudinal axis therebetween, and including:

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an inner frame including an inner-frame tubular portion that circumscribes the central longitudinal axis, has an upstream end and a downstream end, and defines a channel therebetween, the inner frame defining a plurality of inner-frame couplings disposed circumferentially at a longitudinal location of the inner frame,

an outer frame including an outer-frame tubular portion that coaxially circumscribes at least a portion of the inner-frame tubular portion, the outer frame defining a plurality of outer-frame couplings disposed circumferentially at a longitudinal location of the outer frame, and

a plurality of connectors, each connector connecting a respective inner-frame coupling to a respective outer-frame coupling;

a liner, disposed over at least part of the inner-frame tubular portion; and

a plurality of prosthetic leaflets, coupled to the inner-frame tubular portion and disposed within the channel,

and:

the frame assembly: (i) is compressible by a radially-compressive force into a compressed state in which the inner frame is in a compressed state thereof and the outer frame is in a compressed state thereof, (ii) is configured, upon removal of the radially-compressive force, to automatically expand into an expanded state thereof in which the inner frame is in an expanded state thereof and the outer frame is in an expanded state thereof,

in the expanded state of the frame assembly, the prosthetic leaflets are configured to facilitate one-way fluid flow, in a downstream direction, through the channel, and

the connection of the inner-frame couplings to the respective outer-frame couplings is such that expansion of the frame assembly from the compressed state to the expanded state causes the inner-frame tubular portion to slide longitudinally in a downstream direction with respect to the outer-frame tubular portion.

There is further provided, in accordance with an application of the present invention, apparatus for use with a native valve disposed between an atrium and a ventricle of a heart of a subject, the apparatus including:

a tubular portion, having an upstream portion that includes an upstream end, and a downstream portion that includes a downstream end, and shaped to define a lumen between the upstream portion and the downstream portion;

a plurality of prosthetic leaflets, disposed within the lumen, and arranged to provide unidirectional flow of blood from the upstream portion to the downstream portion;

an annular upstream support portion:

having an inner portion that extends radially outward from the upstream portion, and

including one or more fabric pockets disposed circumferentially around the inner portion, each pocket of the one or more pockets having an opening that faces a downstream direction.

In an application:

the upstream support portion includes (i) a plurality of arms that extend radially outward from the tubular portion, and (ii) a covering, disposed over the plurality of arms,

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each arm has (i) a radially-inner part at the inner portion of the upstream support portion, and (ii) a radially-outer part at the outer portion of the upstream support portion,

at the inner portion of the upstream support portion, the covering is closely-fitted between the radially-inner parts of the arms, and

at the outer portion of the upstream support portion, the pockets are formed by the covering being loosely-fitted between the radially-outer parts of the arms.

In an application:

the upstream support portion includes (i) a plurality of arms that extend radially outward from the tubular portion, and (ii) a covering, disposed over the plurality of arms,

each arm has (i) a radially-inner part at the inner portion of the upstream support portion, and (ii) a radially-outer part at the outer portion of the upstream support portion, the radially-outer part being more flexible than the radially-inner part.

In an application:

the upstream support portion includes (i) a plurality of arms that extend radially outward from the tubular portion, and (ii) a covering, disposed over the plurality of arms,

each arm has (i) a radially-inner part at the inner portion of the upstream support portion, and (ii) a radially-outer part at the outer portion of the upstream support portion,

at the outer portion of the upstream support portion, the pockets are formed by each arm curving to form a hook shape.

In an application, each pocket is shaped and arranged to billow in response to perivalvular flow of blood in an upstream direction.

In an application, the apparatus is configured to be trans-luminally delivered to the heart, and implanted at the native valve by expansion of the apparatus, such that the upstream support portion is disposed in the atrium and the tubular portion extends from the upstream support portion into the ventricle, and each pocket is shaped and arranged such that perivalvular flow of blood in an upstream direction presses the pocket against tissue of the atrium.

There is further provided, in accordance with an application of the present invention, apparatus including:

a plurality of prosthetic valve leaflets; and

a frame assembly, including:

a tubular portion defined by a repeating pattern of cells, the tubular portion extending circumferentially around a longitudinal axis so as to define a longitudinal lumen, the prosthetic valve leaflets coupled to the inner frame and disposed within the lumen;

an outer frame, including a plurality of legs, distributed circumferentially around the tubular portion, each leg having a tissue-engaging flange;

an upstream support portion that includes a plurality of arms that extend radially outward from the tubular portion; and

a plurality of appendages, each having a first end that defines a coupling element via which the tubular portion is coupled to the outer frame, and a second end;

and the frame assembly defines a plurality of hubs, distributed circumferentially around the longitudinal axis on a plane that is transverse to the longitudinal axis, each hub defined by convergence and connection of, (i) two adjacent cells of the tubular portion, (ii) an

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arm of the plurality of arms, and (iii) an appendage of the plurality of appendages.

In an application, each hub has six radiating spokes, two of the six spokes being part of a first cell of the two adjacent cells, two of the six spokes being part of a second cell of the two adjacent cells, one of the six spokes being the arm, and one of the six spokes being the second end of the appendage.

In an application, the appendages are in-plane with the tubular portion.

In an application, the appendages are in-plane with the outer frame.

There is further provided, in accordance with an application of the present invention, a method for use with a native valve of a heart of a subject, the method including:

percutaneously advancing to heart, an implant:

including a valve frame, a valve member disposed within a lumen defined by the valve frame, and at least one leg, coupled to the valve frame at a coupling point, and

having an upstream end, a downstream end, and a central longitudinal axis therebetween;

positioning the implant within the heart such that a tissue-engaging flange of the leg is disposed downstream of the valve, and thereafter causing the flange to protrude radially outward from the axis;

subsequently, while an upstream support portion of the valve frame is disposed upstream of the valve, causing the upstream support portion to protrude radially outward from the axis, such that tissue of the valve is disposed between the upstream support portion and the flange; and

subsequently, sandwiching the tissue between the upstream support portion and the flange by reducing a distance between the upstream support portion and the flange by causing longitudinal movement (i) of the upstream support portion toward the coupling point, and (ii) of the tissue-engaging flange away from the coupling point.

In an application, causing the longitudinal movement (i) of the upstream support portion toward the coupling point, and (ii) of the tissue-engaging flange away from the coupling point, includes causing the longitudinal movement by increasing a diameter of the lumen.

There is further provided, in accordance with an application of the present invention, a method for use with a native valve of a heart of a subject, the method including:

percutaneously advancing to heart, an implant:

including a valve frame, a valve member disposed within a lumen defined by the valve frame, and at least one leg, coupled to the valve frame at a coupling point, and

having an upstream end, a downstream end, and a central longitudinal axis therebetween;

positioning the implant within the heart such that an upstream support portion of the valve frame is disposed upstream of the valve, and thereafter causing the upstream support portion to protrude radially outward from the axis;

subsequently, while a tissue-engaging flange of the leg is disposed downstream of the valve, causing the tissue-engaging flange to protrude radially outward from the axis, such that tissue of the valve is disposed between the upstream support portion and the flange; and

subsequently, sandwiching the tissue between the upstream support portion and the flange by reducing a distance between the upstream support portion and the flange by causing longitudinal movement (i) of the

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upstream support portion toward the coupling point, and (ii) of the tissue-engaging flange away from the coupling point.

In an application, causing the longitudinal movement (i) of the upstream support portion toward the coupling point, and (ii) of the tissue-engaging flange away from the coupling point, includes causing the longitudinal movement by increasing a diameter of the lumen.

There is further provided, in accordance with an application of the present invention, a method for use with a native valve of a heart of a subject, the method including:

percutaneously advancing an implant to the heart, the implant:

having an upstream end, a downstream end, and a central longitudinal axis therebetween, and including a tubular portion, an upstream support portion, and a plurality of tissue-engaging flanges;

positioning the implant within the heart such that the upstream support portion is disposed upstream of the valve,

positioning the implant within the heart such that the tissue-engaging flanges are disposed downstream of the valve,

without increasing a diameter of the tubular portion:

causing the upstream support portion to extend radially outward from the axis so as to have a first support-portion span, and

causing the flanges to extend radially outward from the axis so as to have a first flange span; and

subsequently, causing the upstream support portion and the flanges move toward each other by at least 5 mm by increasing a diameter of the tubular portion such that: the upstream support portion extends radially outward so as to have a second support-portion span, the first support-portion span being at least 40 percent as great as the second support-portion span, and the flanges extend radially outward so as to have a second flange span, the first flange span being at least 30 percent as great as the second flange span.

There is further provided, in accordance with an application of the present invention, a method for use with a native valve of a heart of a subject, the method including:

percutaneously advancing an implant to the heart, the implant:

having an upstream end, a downstream end, and a central longitudinal axis therebetween, and including a tubular portion, an upstream support portion, and a plurality of tissue-engaging flanges;

positioning the implant within the heart such that the upstream support portion is disposed upstream of the valve,

positioning the implant within the heart such that the tissue-engaging flanges are disposed downstream of the valve,

without increasing a diameter of the tubular portion:

causing the upstream support portion to extend radially outward from the axis, and

causing the flanges to extend radially outward from the axis so as to have a first flange span; and

subsequently, by increasing a diameter of the tubular portion:

causing the upstream support portion and the flanges move toward each other by at least 5 mm,

causing the upstream support portion to move further radially outward from the axis, and

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causing each flange of the plurality of flanges to translate radially outward so as to have a second flange span that is greater than the first flange span.

The present invention will be more fully understood from the following detailed description of applications thereof, taken together with the drawings, in which:

BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. 1A-B and 2A-H are schematic illustrations of an implant for use with a native valve of a heart of a subject, in accordance with some applications of the invention;

FIGS. 3A-C are schematic illustrations that show structural changes in a frame assembly during transitioning of the assembly between its compressed and expanded states, in accordance with some applications of the invention;

FIGS. 4A-F are schematic illustrations of implantation of the implant at the native valve, in accordance with some applications of the invention;

FIG. 5 is a schematic illustration of a step in the implantation of the implant, in accordance with some applications of the invention;

FIG. 6 is a schematic illustration of the implant, in accordance with some applications of the invention;

FIGS. 7A-B and 8A-B are schematic illustrations of frame assemblies of respective implants, in accordance with some applications of the invention; and

FIGS. 9A-C are schematic illustrations of an implant comprising a frame assembly, in accordance with some applications of the invention.

DETAILED DESCRIPTION OF EMBODIMENTS

Reference is made to FIGS. 1A-B and 2A-H, which are schematic illustrations of an implant 20 for use with a native valve of a heart of a subject, in accordance with some applications of the invention. Implant 20 comprises a frame assembly 22 that has an upstream end 24, a downstream end 26, and a central longitudinal axis ax1 therebetween. Frame assembly 22 comprises a valve frame 30 that comprises a tubular portion 32 that has an upstream end 34 and a downstream end 36, and is shaped to define a lumen 38 through the tubular portion from the upstream end to the downstream end. Tubular portion 32 circumscribes axis ax1, and thereby defines lumen 38 along the axis. Valve frame 30 further comprises an upstream support portion 40, extending from upstream end 34 of tubular portion 32. Frame assembly 22 further comprises at least one leg 50, coupled to valve frame 30 at (e.g., via) a coupling point 52, and having a tissue-engaging flange 54.

Typically, and as described hereinbelow, leg 50 is part of an outer frame (or "leg frame") 60, and frames 30 and 60 define respective coupling elements 31 and 61, which are fixed with respect to each other at coupling points 52. Typically, frames 30 and 60 are coupled to each other only at coupling points 52 (e.g., only via the fixation of coupling elements 31 and 61 with respect to each other).

Implant 20 further comprises a valve member 58 (e.g., one or more prosthetic leaflets) disposed within lumen 38, and configured to facilitate one-way liquid flow through the lumen from upstream end 34 to downstream end 36 (e.g., thereby defining the orientation of the upstream and downstream ends of tubular portion 32). FIG. 1A shows implant 20 in a fully-expanded state, in which frame assembly 22 is in a fully-expanded state. FIG. 1B shows an exploded view of frame assembly 22 in its fully-expanded state. FIGS. 2A-E show respective states of implant 20, which will be

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discussed in more detail hereinbelow with respect to the implantation of the implant and the anatomy in which the implant is implanted. FIG. 2A shows implant 20 in a compressed state (in which frame assembly 22 is in a compressed state), for percutaneous delivery of the implant to the heart of the subject. Typically, in the compressed state, leg 50 (including flange 54 thereof) is in a constrained-flange state in which the flange is generally parallel with axis ax1. Further typically, in the compressed state, upstream support portion 40 is generally tubular, collinear with tubular portion 32 (e.g., extending collinearly from the tubular portion), and disposed around axis ax1.

FIG. 2B shows a state of implant 20 in which tissue-engaging flange 54 of each leg 50 extends radially away from axis ax1 (e.g., radially away from tubular portion 32). FIG. 2C shows a state of implant 20 in which upstream-support portion 40 extends radially away from axis ax1 (and thereby radially away from tubular portion 32). FIG. 2D shows a state of implant 20 in which both flange 54 and portion 40 extend away from axis ax1. In the fully-expanded state (FIGS. 1A-B) both upstream support portion 40 and flange 54 extend radially away from axis ax1. Typically, frame assembly 22 is biased (e.g., shape-set) to assume its fully-expanded state, which is shown in FIG. 2E. Transitioning of implant 20 between the respective states is typically controlled by delivery apparatus, such as by constraining the implant in a compressed state within a delivery tube and/or against a control rod, and selectively releasing portions of the implant to allow them to expand.

In the compressed state of frame assembly 22, tubular portion 32 has a diameter d1, and in the expanded state, the tubular portion has a diameter d2 that is greater than diameter d1. For some applications, diameter d1 is 4-15 mm, (e.g., 5-11 mm) and diameter d2 is 20-50 mm, (e.g., 23-33 mm). Frame assembly 22 is configured such that increasing the diameter of tubular portion 32 (e.g., from d1 to d2) causes longitudinal movement of flange 54 away from coupling point 52. In the same way, reducing the diameter of tubular portion 32 (e.g., from d2 to d1) causes longitudinal movement of flange 54 toward coupling point 52. It is to be noted that the term "longitudinal movement" (including the specification and the claims) means movement parallel with central longitudinal axis ax1. Therefore, longitudinal movement of flange 54 away from coupling point 52 means increasing a distance, measured parallel with longitudinal axis ax1, between flange 54 and coupling point 52. An example of such a configuration is described in more detail with respect to FIG. 3A.

Thus, expansion of tubular portion 32 from its compressed state toward its expanded state (i) increases a circumferential distance between each of coupling points 52 and its adjacent coupling points (e.g., between each of outer-frame coupling elements 61 and its adjacent outer-frame coupling elements) (e.g., from d8 to d9), and (ii) moves legs 50 in a longitudinally upstream direction with respect to the tubular portion.

Typically, frame assembly 22 is configured such that increasing the diameter of tubular portion 32 also causes longitudinal movement of upstream support portion 40 toward coupling point 52, e.g., as described in more detail with respect to FIGS. 3B-C. Typically, frame assembly 22 is configured such that increasing the diameter of tubular portion 32 also causes longitudinal movement of upstream end 34 of tubular portion 32 toward coupling point 52. In the same way, reducing the diameter of tubular portion 32 causes longitudinal movement of upstream end 34 away from coupling point 52.

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For some applications, upstream support portion 40 comprises a plurality of arms 46 that each extends radially outward from tubular portion 32 (e.g., from upstream end 34 of the tubular portion). Arms 46 are typically flexible. For some such applications, arms 46 are coupled to tubular portion 32 such that each arm may deflect independently of adjacent arms during implantation (e.g., due to anatomical topography).

For some applications, upstream support portion 40 comprises a plurality of barbs 48 that extend out of a downstream surface of the upstream support portion. For example, each arm 46 may comprise one or more of barbs 48. Barbs 48 press into tissue upstream of the native valve (e.g., into the valve annulus), thereby inhibiting downstream movement of implant 20 (in addition to inhibition of downstream movement provided by the geometry of upstream support portion 40).

One or more surfaces of frame assembly 22 are covered with a covering 23, which typically comprises a flexible sheet, such as a fabric, e.g., comprising polyester. Typically, covering 23 covers at least part of tubular portion 32, typically lining an inner surface of the tubular portion, and thereby defining lumen 38.

Further typically, upstream support portion 40 is covered with covering 23, e.g., extending between arms 46 to form an annular shape. It is hypothesized that this reduces a likelihood of paravalvular leakage. For such applications, excess covering 23 may be provided between arms 46 of upstream support portion 40, so as to facilitate their independent movement. Although FIG. 1A shows covering 23 covering an upstream side of upstream support portion 40, the covering typically additionally (or alternatively) covers the downstream side of the upstream support portion. For example, covering 23 may extend over the tips of arms 46 and down the outside of the arms, or a separate piece of covering may be provided on the downstream side of the upstream support portion.

Alternatively, each arm 46 may be individually covered in a sleeve of covering 23, thereby facilitating independent movement of the arms.

For some applications, at least part of legs 50 (e.g., flanges thereof) is covered with covering 23.

Typically, frame assembly 22 comprises a plurality of legs 50 (e.g., two or more legs, e.g., 2-16 legs, such as 4-12 legs, such as 6-12 legs), arranged circumferentially around valve frame 30 (e.g., around the outside of tubular portion 32). Typically, frame assembly 22 comprises a plurality of coupling points 52 at which the legs are coupled to valve frame 30.

As described in more detail hereinbelow (e.g., with reference to FIG. 3A), each leg 50 is typically coupled to a coupling point 52 via a strut 70. For some applications, each leg 50 is coupled to a plurality of (e.g., two) coupling points 52 via a respective plurality of (e.g., two) struts 70. For some such applications, frame assembly 22 is arranged such that, in the expanded state of the frame assembly, leg 50 is disposed, circumferentially with respect to tubular portion 32, between two struts, and each of the two struts are disposed, circumferentially with respect to the tubular portion, between the leg and a respective coupling point 52.

For some applications, a plurality of (e.g., two) legs are coupled to each coupling point 52 via a respective plurality of (e.g., two) struts 70. For some such applications, frame assembly 22 is arranged such that, in the expanded state of the frame assembly, coupling point 52 is disposed, circumferentially with respect to tubular portion 32, between two struts 70, and each of the two struts are disposed, circum-

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ferentially with respect to the tubular portion, between the coupling point and a respective leg 50.

For some applications, frame assembly 22 comprises an outer frame (e.g., a leg frame) 60 that circumscribes tubular portion 32, comprises (or defines) the plurality of legs 50 and the plurality of struts 70, and is coupled to valve frame 30 at the plurality of coupling points 52, such that the plurality of legs is distributed circumferentially around the tubular portion. For such applications, outer frame 60 comprises a ring 66 that is defined by a pattern of alternating peaks 64 and troughs 62, and that typically circumscribes tubular portion 32. For example, the ring may comprise struts 70, extending between the peaks and troughs. Peaks 64 are longitudinally closer to upstream end 34 of tubular portion 32 than to downstream end 36, and troughs 62 are longitudinally closer to the downstream end than to the upstream end. (It is to be noted that throughout this patent application, including the specification and the claims, the term "longitudinally" means with respect to longitudinal axis ax1. For example, "longitudinally closer" means closer along axis ax1 (whether positioned on axis ax1 or lateral to axis ax1), and "longitudinal movement" means a change in position along axis ax1 (which may be in addition to movement toward or away from axis ax1).) Therefore, peaks 64 are closer than troughs 62 to upstream end 34, and troughs 62 are closer than peaks 64 to downstream end 36. For applications in which frame 60 comprises ring 66, each leg 50 is coupled to the ring (or defined by frame 60) at a respective trough 62.

In the embodiment shown, the peaks and troughs are defined by ring 66 having a generally zig-zag shape. However, the scope of the invention includes ring 66 having another shape that defines peaks and troughs, such as a serpentine or sinusoid shape.

For applications in which frame assembly 22 has a plurality of coupling points 52, the coupling points (and therefore coupling elements 31 and 61) are disposed circumferentially around the frame assembly (e.g., around axis ax1), typically on a transverse plane that is orthogonal to axis ax1. This transverse plane is illustrated by the position of section A-A in FIG. 2B. Alternatively, coupling points 52 may be disposed at different longitudinal heights of frame assembly 22, e.g., such that different flanges 54 are positioned and/or moved differently to others. Typically, coupling points 52 (and therefore coupling elements 31 and 61) are disposed longitudinally between upstream end 24 and downstream end 26 of frame assembly 22, but not at either of these ends. Further typically, coupling points 52 are disposed longitudinally between upstream end 34 and downstream end 36 of tubular portion 32, but not at either of these ends. For example, the coupling points may be more than 3 mm (e.g., 4-10 mm) both from end 34 and from end 36. It is hypothesized that this advantageously positions the coupling points at a part of tubular portion 32 that is more rigid than end 34 or end 36.

It is to be noted that leg 50 is typically expandable into its expanded state (e.g., a released-flange state) such that flange 54 extends away from axis ax1, independently of increasing the diameter of tubular portion 32 (e.g., as shown in FIGS. 2B & 2D). Similarly, upstream support portion 40 is typically expandable into its expanded state (e.g., a released-arm state) such that it (e.g., arms 46 thereof) extends away from axis ax1, independently of increasing the diameter of tubular portion 32 (e.g., as shown in FIGS. 2C & 2D). The state shown in FIG. 2D may be considered to be an intermediate state. Therefore, implant 20 is typically configured such that legs 50 (e.g., flanges 54 thereof) and upstream support

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portion 40 are expandable such that they both extend away from axis ax1, while retaining a distance d3 therebetween. This distance is subsequently reducible to a distance d4 by expanding tubular portion 32 (e.g., shown in FIG. 2E).

For some applications, while tubular portion 32 remains in its compressed state, flange 54 can extend away from axis ax1 over 40 percent (e.g., 40-80 percent, such as 40-70 percent) of the distance that it extends from the axis subsequent to the expansion of the tubular portion. For example, for applications in which implant 20 comprises a flange on opposing sides of the implant, a span d15 of the flanges while tubular portion 32 is in its compressed state may be at least 40 percent (e.g., 40-80 percent, such as 40-70 percent) as great as a span d16 of the flanges subsequent to the expansion of the tubular portion. For some applications, span d15 is greater than 15 mm and/or less than 50 mm (e.g., 20-30 mm). For some applications, span d16 is greater than 30 mm and/or less than 60 mm (e.g., 40-50 mm). It is to be noted that flange 54 is effectively fully expanded, with respect to other portions of leg 50 and/or with respect to tubular portion 32, before and after the expansion of the tubular portion.

Similarly, for some applications, while tubular portion 32 remains in its compressed state, upstream support portion 40 (e.g., arms 46) can extend away from axis ax1 over 30 percent (e.g., 30-70 percent) of the distance that it extends from the axis subsequent to the expansion of the tubular portion. That is, for some applications, a span d17 of the upstream support portion while tubular portion 32 is in its compressed state may be at least 30 percent (e.g., 30-70 percent) as great as a span d18 of the upstream support portion subsequent to the expansion of the tubular portion. For some applications, span d17 is greater than 16 mm (e.g., greater than 20 mm) and/or less than 50 mm (e.g., 30-40 mm). For some applications, span d18 is greater than 40 mm and/or less than 65 mm (e.g., 45-56 mm, such as 45-50 mm). It is to be noted that upstream support portion 40 is effectively fully expanded, with respect to tubular portion 32, before and after the expansion of the tubular portion.

It is to be noted that when tubular portion 32 is expanded, flanges 54 typically translate radially outward from span d15 to span d16 (e.g., without deflecting). Typically, upstream support portion 40 behaves similarly (e.g., arms 46 translated radially outward from span d17 to span d18, e.g., without deflecting). That is, an orientation of each flange 54 and/or each arm 46 with respect to tubular portion 32 and/or axis ax1 is typically the same in the state shown in FIG. 2D as it is in the state shown in FIG. 2E. Similarly, for some applications an orientation of each flange 54 with respect to upstream support portion 40 (e.g., with respect to one or more arms 46 thereof) is the same before and after expansion of tubular portion 32.

For some applications, increasing the diameter of tubular portion 32 from d1 to d2 causes greater than 1 mm and/or less than 20 mm (e.g., 1-20 mm, such as 1-10 mm or 5-20 mm) of longitudinal movement of flange 54 away from coupling point 52. For some applications, increasing the diameter of tubular portion 32 from d1 to d2 causes greater than 1 mm and/or less than 20 mm (e.g., 1-20 mm, such as 1-10 mm or 5-20 mm) of longitudinal movement of upstream support portion 40 toward coupling point 52. For some applications, distance d3 is 7-30 mm. For some applications, distance d4 is 0-15 mm (e.g., 2-15 mm). For some applications, increasing the diameter of tubular portion 32 from d1 to d2 reduces the distance between the upstream support portion and flanges 54 by more than 5 mm and/or less than 30 mm, such as 5-30 mm (e.g., 10-30 mm, such as

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10-20 mm or 20-30 mm). For some applications, the difference between d3 and d4 is generally equal to the difference between d1 and d2. For some applications, the difference between d3 and d4 is more than 1.2 and/or less than 3 times (e.g., 1.5-2.5 times, such as about 2 times) greater than the difference between d1 and d2.

For some applications, flanges 54 curve such that a tip of each flange is disposed at a shallower angle with respect to inner region 42 of upstream support portion 40, than are portions of leg 50 that are closer to downstream end 26 of frame assembly 22. For some such applications, a tip of each flange may be generally parallel with inner region 42. For some applications, while tubular portion 32 is in its expanded state, a tip portion 55 of each flange 54 that extends from the tip of the flange at least 2 mm along the flange, is disposed within 2 mm of upstream support portion 40. Thus, for some applications, while tubular portion 32 is in its expanded state, for at least 5 percent (e.g., 5-8 percent, or at least 8 percent) of span 18 of upstream support portion 40, the upstream support portion is disposed within 2 mm of a flange 54.

For some applications, in the absence of any obstruction (such as tissue of the valve or covering 23) between flange 54 and upstream support portion 40, increasing the diameter of tubular portion 32 from d1 to d2 causes the flange and the upstream support portion to move past each other (e.g., the flange may move between arms 46 of the upstream support portion), such that the flange is closer to the upstream end of implant 20 than is the upstream support portion, e.g., as shown hereinbelow for frame assemblies 122 and 222, mutatis mutandis. (For applications in which upstream support portion 40 is covered by covering 23, flanges 54 typically don't pass the covering. For example, in the absence of any obstruction, flanges 54 may pass between arms 46, and press directly against covering 23.) It is hypothesized that for some applications this configuration applies greater force to the valve tissue being sandwiched, and thereby further facilitates anchoring of the implant. That is, for some applications, distance d3 is smaller than the sum of distance d5 and a distance d14 (described with reference to FIG. 3C). For some applications, increasing the diameter of tubular portion 32 from d1 to d2 advantageously causes flanges 54 and upstream support portion 40 to move greater than 3 mm and/or less than 25 mm (e.g., greater than 5 mm and/or less than 15 mm, e.g., 5-10 mm, such as about 7 mm) with respect to each other (e.g., toward each other and then past each other).

For some applications, in the expanded state of frame assembly 22, upstream support portion 40 has an inner region (e.g., an inner ring) 42 that extends radially outward at a first angle with respect to axis ax1 (and typically with respect to tubular portion 32), and an outer region (e.g., an outer ring) 44 that extends, from the inner region, further radially outward from the tubular portion at a second angle with respect to the tubular portion, the second angle being smaller than the first angle. For example, for some applications inner region 42 extends radially outward at an angle α_1 of 60-120 degrees (e.g., 70-110 degrees) with respect to axis ax1, and outer region 44 extends radially outward at an angle α_2 of 5-70 degrees (e.g., 10-60 degrees) with respect to axis ax1.

It is to be noted that angles α_1 and α_2 are measured between the respective region support portion 40, and the portion of axis ax1 that extends in an upstream direction from the level of frame assembly 22 at which the respective region begins to extend radially outward.

For some applications in which implant **20** is configured to be placed at an atrioventricular valve (e.g., a mitral valve or a tricuspid valve) of the subject, region **42** is configured to be placed against the upstream surface of the annulus of the atrioventricular valve, and region **44** is configured to be placed against the walls of the atrium upstream of the valve.

For some applications, each arm **46** defines a rigid portion **43** and a flexible portion **45** that is disposed radially outward from the rigid portion, and is more flexible than the rigid portion. Typically, at least part of flexible portion **45** is disposed in outer region **44**, and at least part of rigid portion **43** is disposed in inner region **42**. Therefore, typically overall, outer region **44** is more flexible than inner region **42**. To achieve this differential flexibility, each arm **46** may have a different structure in portion **45** and/or region **44**, than it has in portion **43** and/or region **42**.

FIG. 2G shows an arm **46a**, which is a variant of arm **46**. Arm **46a** defines flexible portion **45a**, which is a variant of flexible portion **45**. In flexible portion **45a**, the material from which arm **46a** is formed is thinner than in rigid portion **43**, but defines a serpentine or zig-zag pattern that increases the overall thickness of portion **45a**, and as shown, may make the overall thickness of portion **45a** equal to or greater than the overall thickness of portion **43**.

FIG. 2H shows an arm **46b**, which is a variant of arm **46**. Arm **46b** defines flexible portion **45b**, which is a variant of flexible portion **45**. In flexible portion **45b**, arm **46b** defines holes **47** in the material from which arm **46b** is formed. Nonetheless, as shown, the overall thickness of portion **45b** may be equal to or greater than the overall thickness of portion **43**.

Therefore, as shown for arms **46a** and **46b**, flexible portion **45** covers an overall surface area including (i) the area in which the material from which the arm is formed occupies, and (ii) spaces, such as holes, or gaps between turns. For some applications, the material from which the arm is formed occupies less than 80 percent and/or more than 20 percent of the overall surface area (e.g., 20-80 percent, e.g., 40-80 percent, e.g., 40-70 percent of the overall surface area).

It is hypothesized that the relative rigidity of portion **43** and/or region **42** provides resistance against axial movement (e.g., ventricular migration) of implant **20**, while the relative flexibility of portion **45** and/or region **44** facilitates conformation of upstream support portion **40** to the atrial anatomy and/or lateral movement of the implant within the native valve.

For some applications, and as shown (e.g., in FIGS. 1A and 2E-F), in the expanded state of frame assembly **22**, the majority of flexible portion **45** (e.g., more than 60 percent and/or less than 90 percent) is disposed further radially outward than span **d16** of flanges **54**.

For some applications, two or more of arms **46** are connected by a connector (not shown), reducing the flexibility, and/or the independence of movement of the connected arms relative to each other. For some applications, arms **46** are connected in particular sectors of upstream support portion **40**, thereby making these sectors more rigid than sectors in which the arms are not connected. For example, a relatively rigid sector may be provided to be placed against the posterior portion of the mitral annulus, and a relatively flexible sector may be provided to be placed against the anterior side of the mitral annulus, so as to reduce forces applied by upstream support portion **40** on the aortic sinus.

For some applications, and as shown, coupling points **52** are disposed closer to downstream end **26** of frame assembly **22** than are flanges **54**, or is upstream support portion **40**.

As described in more detail with respect to FIGS. 4A-F, the movement of flange **54** away from coupling point **52** (and the typical movement of upstream support portion **40** toward the coupling point) facilitates the sandwiching of tissue of the native valve (e.g., leaflet and/or annulus tissue) between the flange and the upstream support portion, thereby securing implant **20** at the native valve.

Typically, in the compressed state of tubular portion **32**, a downstream end of each leg **50** is longitudinally closer than valve-frame coupling elements **31** to downstream end **36**, and flange **54** of each leg is disposed longitudinally closer than the valve-frame coupling elements to upstream end **34**. Typically, this is also the case in the expanded state of tubular portion **32**.

FIGS. 3A-C show structural changes in frame assembly **22** during transitioning of the assembly between its compressed and expanded states, in accordance with some applications of the invention. FIGS. 3A-C each show a portion of the frame assembly, the structural changes thereof being representative of the structural changes that occur in other portions of the frame assembly. FIG. 3A shows a leg **50** and struts **70** (e.g., a portion of outer frame **60**), and illustrates the structural changes that occur around outer frame **60**. FIG. 3B shows a portion of valve frame **30**, and illustrates the structural changes that occur around the valve frame. FIG. 3C shows valve frame **30** as a whole. In each of FIGS. 3A-C, state (A) illustrates the structure while frame assembly **22** (and in particular tubular portion **32**) is in its compressed state, and state (B) illustrates the structure while the frame assembly (and in particular tubular portion **32**) is in its expanded state.

FIG. 3A shows structural changes in the coupling of legs **50** to coupling point **52** (e.g., structural changes of outer frame **60**) during the transitioning of frame assembly **22** (and in particular tubular portion **32**) between its compressed and expanded states. Each leg **50** is coupled to valve frame **30** via at least one strut **70**, which connects the leg to coupling point **52**. Typically, each leg **50** is coupled to valve frame **30** via a plurality of struts **70**. A first end **72** of each strut **70** is coupled to leg **50**, and a second end **74** of each strut is coupled to a coupling point **52**. As described hereinabove, for applications in which frame **60** comprises ring **66**, each leg **50** is coupled to the ring at a respective trough **62**. Ring **66** may comprise struts **70**, extending between the peaks and troughs, with each first end **72** at (or close to) a trough **62**, and each second end **74** at (or close to) a peak **64**.

In the compressed state of frame assembly **22** (and in particular of tubular portion **32**), each strut **70** is disposed at a first angle in which first end **72** is disposed closer than second end **74** to the downstream end of the frame assembly. Expansion of frame assembly **22** (and in particular of tubular portion **32**) toward its expanded state causes strut **70** to deflect to a second angle. This deflection moves first end **72** away from the downstream end of frame assembly **22**. That is, in the expanded state of frame assembly **22**, first end **72** is further from the downstream end of the frame assembly than it is when the frame assembly is in its compressed state. This movement is shown as a distance **d5** between the position of end **72** in state (A) and its position in state (B). This movement causes the above-described movement of flanges **54** away from coupling points **52**. As shown, flanges **54** typically move the same distance **d5** in response to expansion of frame assembly **22**.

For applications in which outer frame 60 comprises ring 66, the pattern of alternating peaks and troughs may be described as having an amplitude longitudinally between the peaks and troughs, i.e., measured parallel with central longitudinal axis ax1 of frame assembly 22, and the transition between the compressed and expanded states may be described as follows: In the compressed state of frame assembly 22 (and in particular of tubular portion 32), the pattern of ring 66 has an amplitude d20. In the expanded state frame assembly 22 (and in particular of tubular portion 32), the pattern of ring 66 has an amplitude d21 that is lower than amplitude d20. Because (i) it is at peaks 64 that ring 66 is coupled to valve frame 30 at coupling points 52, and (ii) it is at troughs 62 that ring 66 is coupled to legs 50, this reduction in the amplitude of the pattern of ring 66 moves legs 50 (e.g., flanges 54 thereof) longitudinally further from the downstream end of the frame assembly. The magnitude of this longitudinal movement (e.g., the difference between magnitudes d20 and d21) is equal to d5.

Typically, distance d5 is the same distance as the distance that flange 54 moves away from coupling point 52 during expansion of the frame assembly. That is, a distance between flange 54 and the portion of leg 50 that is coupled to strut 70, typically remains constant during expansion of the frame assembly. For some applications, the longitudinal movement of flange 54 away from coupling point 52 is a translational movement (e.g., a movement that does not include rotation or deflection of the flange).

For some applications, a distance d6, measured parallel to axis ax1 of frame assembly 22, between coupling point 52 and first end 72 of strut 70 while assembly 22 is in its compressed state, is 3-15 mm. For some applications, a distance d7, measured parallel to axis ax1, between coupling point 52 and first end 72 of strut 70 while assembly 22 is in its expanded state, is 1-5 mm (e.g., 1-4 mm).

For some applications, amplitude d20 is 2-10 mm (e.g., 4-7 mm). For some applications, amplitude d21 is 4-9 mm (e.g., 5-7 mm).

For some applications, and as shown, in the expanded state, first end 72 of strut 70 is disposed closer to the downstream end of frame assembly 22 than is coupling point 52. For some applications, in the expanded state, first end 72 of strut 70 is disposed further from the downstream end of frame assembly 22 than is coupling point 52.

For applications in which frame assembly 22 comprises a plurality of legs 50 and a plurality of coupling points 52 (e.g., for applications in which the frame assembly comprises outer frame 60) expansion of the frame assembly increases a circumferential distance between adjacent coupling points 52, and an increase in a circumferential distance between adjacent legs 50. FIG. 3A shows such an increase in the circumferential distance between adjacent coupling points 52, from a circumferential distance d8 in the compressed state to a circumferential distance d9 in the expanded state. For some applications, distance d8 is 1-6 mm. For some applications, distance d9 is 3-15 mm.

For some applications, in addition to being coupled via ring 66 (e.g., struts 70 thereof) legs 50 are also connected to each other via connectors 78. Connectors 78 allow the described movement of legs 50 during expansion of frame assembly 22, but typically stabilize legs 50 relative to each other while the frame assembly is in its expanded state. For example, connectors 78 may bend and/or deflect during expansion of the frame assembly.

FIGS. 3B-C show structural changes in valve frame 30 during the transitioning of frame assembly 22 between its compressed and expanded states. Tubular portion 32 of

valve frame 30 is defined by a plurality of cells 80, which are defined by the repeating pattern of the valve frame. When frame assembly 22 is expanded from its compressed state toward its expanded state, cells 80 (i) widen from a width d10 to a width d11 (measured orthogonal to axis ax1 of the frame assembly), and (ii) shorten from a height d12 to a height d13 (measured parallel to axis ax1 of the frame assembly). This shortening reduces the overall height (i.e., a longitudinal length between upstream end 34 and downstream end 36) of tubular portion 32 from a height d22 to a height d23, and thereby causes the above-described longitudinal movement of upstream support portion 40 toward coupling points 52 by a distance d14 (shown in FIG. 3C). For some applications, and as shown, coupling points 52 are disposed at the widest part of each cell.

Due to the configurations described herein, the distance by which flanges 54 move with respect to (e.g., toward, or toward-and-beyond) upstream support portion 40 (e.g., arms 46 thereof), is typically greater than the reduction in the overall height of tubular portion 32 (e.g., more than 20 percent greater, such as more than 30 percent greater, such as more than 40 percent greater). That is, implant 20 comprises:

- a valve frame (30) that comprises a tubular portion (32) that circumscribes a longitudinal axis (ax1) of the valve frame so as to define a lumen (38) along the axis, the tubular portion having an upstream end (34), a downstream end (36), a longitudinal length therebetween, and a diameter (e.g., d1 or d2) transverse to the longitudinal axis;

- a valve member (58), coupled to the tubular portion, disposed within the lumen, and arranged to provide unidirectional upstream-to-downstream flow of blood through the lumen;

- an upstream support portion (40), coupled to the tubular portion; and

- an outer frame (60), coupled to the tubular portion, and comprising a tissue-engaging flange (54),

wherein:

- the implant has a first state (e.g., as shown in FIG. 2D and FIG. 4D) and a second state (e.g., as shown in FIG. 2E and FIG. 4E),

- in both the first state and the second state, (i) the upstream support portion extends radially outward from the tubular portion, and (ii) the tissue-engaging flange extends radially outward from the tubular portion, and the tubular portion, the upstream support portion, and the outer frame are arranged such that transitioning of the implant from the first state toward the second state:

- increases the diameter of the tubular portion by a diameter-increase amount (e.g., the difference between d1 and d2),

- decreases the length of the tubular portion by a length-decrease amount (e.g., the difference between d22 and d23), and

- moves the flange a longitudinal distance with respect to (e.g., toward or toward-and-beyond) the upstream support portion (e.g., the difference between d3 and d4), this distance being greater than the length-decrease amount.

As shown in the figures, valve frame 30 is typically coupled to outer frame 60 by coupling between (i) a valve-frame coupling element 31 defined by valve frame 30, and (ii) an outer-frame coupling element 61 defined by outer frame 60 (e.g., an outer-frame coupling element is coupled to end 74 of each strut). Typically, elements 31 and 61 are fixed with respect to each other. Each coupling point 52 is

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thereby typically defined as the point at which a valve-frame coupling element and a corresponding outer-frame coupling element **61** are coupled (e.g., are fixed with respect to each other). For some applications, and as shown, elements **31** and **61** are eyelets configured to be coupled together by a connector, such as a pin or suture. Alternatively or additionally, elements **31** and **61** are soldered or welded together. For example, elements **31** and **61** may be coupled together by a pin that passes through the eyelets, and is welded in place.

Typically, and as shown, valve-frame coupling elements **31** are defined by tubular portion **32**, and are disposed circumferentially around central longitudinal axis **ax1**. Outer-frame coupling elements **61** are coupled to ring **66** (or defined by frame **60**, such as by ring **66**) at respective peaks **64**.

As shown (e.g., in FIGS. 2A-E), valve frame **30** (e.g., tubular portion **32** thereof) and outer frame **60** (e.g., ring **66** thereof) are arranged in a close-fitting coaxial arrangement, in both the expanded and compressed states of frame assembly **22**. Ignoring spaces due to the cellular structure of the frames, a radial gap **d19** between valve frame **30** (e.g., tubular portion **32** thereof) and outer frame **60** (e.g., ring **66** thereof) is typically less than 2 mm (e.g., less than 1 mm), in both the compressed and expanded states, and during the transition therebetween. This is facilitated by the coupling between frames **30** and **60**, and the behavior, described hereinabove, of frame **60** in response to changes in the diameter of tubular portion **32** (e.g., rather than solely due to delivery techniques and/or tools). For some applications, more than 50 percent (e.g., more than 60 percent) of ring **66** is disposed within 2 mm of tubular portion **32** in both the compressed and expanded states, and during the transition therebetween. For some applications, more than 50 percent (e.g., more than 60 percent) of outer frame **60**, except for flanges **54**, is disposed within 2 mm of tubular portion **32** in both the compressed and expanded states, and during the transition therebetween.

The structural changes to frame assembly **22** (e.g., to outer frame **60** thereof) are described hereinabove as they occur during (e.g., as a result of) expansion of the frame assembly (in particular tubular portion **32** thereof). This is the natural way to describe these changes because, as described hereinbelow with respect to FIGS. 4A-6, assembly **22** is in its compressed state during percutaneous delivery to the implant site, and is subsequently expanded. However, the nature of implant **20** may be further understood by describing structural changes that occur during compression of the frame assembly (e.g., a transition from the expanded state in FIG. 2E to the intermediate state in FIG. 2D), in particular tubular portion **32** thereof (including if tubular portion **32** were compressed by application of compressive force to the tubular portion, and not to frame **60** except via the tubular portion pulling frame **60** radially inward). Such descriptions may also be relevant because implant **20** is typically compressed (i.e., "crimped") soon before its percutaneous delivery, and therefore these changes may occur while implant **20** is in the care of the operating physician.

For some applications, the fixation of peaks **64** to respective sites of tubular portion **32** is such that compression of the tubular portion from its expanded state toward its compressed state such that the respective sites of the tubular portion pull the peaks radially inward via radially-inward tension on coupling points **52**: (i) reduces a circumferential distance between each of the coupling points and its adjacent coupling points (e.g., from **d9** to **d8**), and (ii) increases the amplitude of the pattern of ring **66** (e.g., from **d21** to **d20**).

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For some applications, the fixation of outer-frame coupling elements **61** to valve-frame coupling elements **31** is such that compression of tubular portion **32** from its expanded state toward its compressed state such that the valve-frame coupling elements pull the outer-frame coupling elements radially inward: (i) reduces a circumferential distance between each of the outer-frame coupling elements and its adjacent outer-frame coupling elements (e.g., from **d9** to **d8**), and (ii) increases the amplitude of the pattern of ring **66** (e.g., from **d21** to **d20**).

For some applications, the fixation of peaks **64** to the respective sites of tubular portion **32** is such that compression of the tubular portion from its expanded state toward its compressed state (i) pulls the peaks radially inward via radially-inward pulling of the respective sites of the tubular portion on the peaks, (ii) reduces a circumferential distance between each of coupling points **52** and its adjacent coupling points (e.g., from **d9** to **d8**), and (iii) increases the amplitude of the pattern of ring **66** (e.g., from **d21** to **d20**), without increasing radial gap **d19** between valve frame **30** (e.g., tubular portion **32** thereof) and the ring by more than 1.5 mm.

For some applications, the fixation of outer-frame coupling elements **61** with respect to valve-frame coupling elements **31** is such that compression of tubular portion **32** from its expanded state toward its compressed state (i) pulls outer-frame coupling elements **61** radially inward via radially-inward pulling of valve-frame coupling elements **31** on outer-frame coupling elements **61**, (ii) reduces a circumferential distance between each of the outer-frame coupling elements and its adjacent outer-frame coupling elements (e.g., from **d9** to **d8**), and (iii) increases the amplitude of the pattern of ring **66** (e.g., from **d21** to **d20**), without increasing radial gap **d19** between valve frame **30** (e.g., tubular portion **32** thereof) and the ring by more than 1.5 mm.

Reference is made to FIGS. 4A-F, which are schematic illustrations of implantation of implant **20** at a native valve **10** of a heart **4** of a subject, in accordance with some applications of the invention. Valve **10** is shown as a mitral valve of the subject, disposed between a left atrium **6** and a left ventricle **8** of the subject. However, implant **20** may be implanted at another heart valve of the subject, mutatis mutandis. Similarly, although FIGS. 4A-F show implant **20** being delivered transseptally via a sheath **88**, the implant may alternatively be delivered by any other suitable route, such as transatrially, or transapically.

Implant **20** is delivered, in its compressed state, to native valve **10** using a delivery tool **89** that is operable from outside the subject (FIG. 4A). Typically, implant **20** is delivered within a delivery capsule **90** of tool **89**, which retains the implant in its compressed state. Delivery capsule **90** is disposed at a distal end of tool **89**. A transseptal approach, such as a transfemoral approach, is shown. Typically, implant **20** is positioned such that at least flanges **54** are disposed downstream of the native valve (i.e., within ventricle **8**). At this stage, frame assembly **22** of implant **20** is as shown in FIG. 2A.

Subsequently, flanges **54** are allowed to protrude radially outward, as described hereinabove, e.g., by releasing them from capsule **90** (FIG. 4B). For example, and as shown, capsule **90** may comprise a distal capsule-portion **92** and a proximal capsule-portion **94**, and the distal capsule-portion may be moved distally with respect to implant **20**, so as to expose flanges **54**. At this stage, frame assembly **22** of implant **20** is as shown in FIG. 2B.

Subsequently, implant **20** is moved upstream, such that upstream support portion **40**, in its compressed state, is

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disposed upstream of leaflets **12** (i.e., within atrium **6**). For some applications, the upstream movement of implant **20** causes flanges **54** to engage leaflets **12**. However, because of the relatively large distance **d3** provided by implant **20** (described hereinabove), for some applications it is not necessary to move the implant so far upstream that flanges **54** tightly engage leaflets **12** and/or pull the leaflets upstream of the valve annulus. Upstream support portion **40** is then allowed to expand such that it protrudes radially outward, as described hereinabove, e.g., by releasing it from capsule-portion **94** (FIG. **4D**). For example, and as shown, proximal capsule-portion **94** may be moved proximally with respect to implant **20**, so as to expose upstream support portion **40**. At this stage, frame assembly **22** of implant **20** is as shown in FIG. **2D**, in which: (i) distance **d3** exists between upstream support portion **40** and flanges **54**, (ii) the flanges have span **d15**, (iii) the upstream support portion has span **d17**, and (iv) tubular portion **32** has diameter **d1**.

Typically, expansion of frame assembly **22** is inhibited by distal capsule-portion **92**, e.g., by inhibiting expansion of tubular portion **32**. For example, a downstream region **33** of frame assembly **22**, downstream of flanges **54**, may remain compressed within distal capsule-portion **92**. Alternatively or additionally, the expansion may be inhibited by another portion of delivery tool **89** (e.g., a portion of the delivery tool that is disposed within lumen **38**).

Subsequently, implant **20** is allowed to expand toward its expanded state, such that tubular portion **32** widens to diameter **d2**, and the distance between upstream support portion **40** and flanges **54** reduces to distance **d4** (FIG. **4E**). This is typically achieved by deploying downstream region **33** from capsule-portion **92**. This expansion sandwiches tissue of valve **10** (typically including annular tissue and/or leaflets **12**) between upstream support portion **40** and flanges **54**, thereby securing implant **20** at the valve. FIG. **4F** shows delivery capsule **90** having been removed from the body of the subject, leaving implant **20** in place at valve **10**.

As described hereinabove, implant **20** is configured such that when tubular portion **32** is expanded, flanges **54** and upstream support portion **40** move a relatively large distance toward each other. This enables distance **d3** to be relatively large, while distance **d4** is sufficiently small to provide effective anchoring. As also described hereinabove, implant **20** is configured such that flanges **54** and upstream support portion **40** can extend radially outward a relatively large distance while tubular portion **32** remains compressed. It is hypothesized that for some applications, these configurations (independently and/or together) facilitate effective anchoring of implant **20**, by facilitating placement of a relatively large proportion of valve tissue (e.g., leaflets **12**) between the flanges and the upstream support portion prior to expanding tubular portion **32** and sandwiching the valve tissue.

It is further hypothesized that the relatively great radially-outward extension of flanges **54** and upstream support portion **40** prior to expansion of tubular portion **32**, further facilitates the anchoring/sandwiching step by reducing radially-outward pushing of the valve tissue (e.g., leaflets **12**) during the expansion of the tubular portion, and thereby increasing the amount of valve tissue that is sandwiched.

It is yet further hypothesized that this configuration of implant **20** facilitates identifying correct positioning of the implant (i.e., with upstream support portion **40** upstream of leaflets **12** and flanges **54** downstream of the leaflets) prior to expanding tubular portion **32** and sandwiching the valve tissue.

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It is to be noted that the deployment of implant **20** is therefore achieved by:

unhousing a longitudinally-intermediate portion of the implant (to allow flanges **54** to expand), the longitudinally-intermediate portion being longitudinally between a longitudinally-proximal portion of the implant and a longitudinally distal portion of the implant;

subsequently, unhousing the longitudinally-proximal portion of the implant (to allow upstream support portion **40** to expand); and

subsequently, unhousing the longitudinally-distal portion of the implant (to allow the longitudinally-distal portion of the implant, typically including the tubular portion within which valve member **58** is disposed, to expand).

It is to be further noted that the deployment of implant **20** may be described with respect to delivery tool **89**. For example:

distal capsule-portion **92** is moved distally with respect to the implant (to allow flanges **54** to expand);

subsequently, proximal capsule-portion **94** is moved proximally with respect to the implant (to allow upstream support portion **40** to expand); and

subsequently, distal capsule-portion **92** is again moved distally with respect to the implant (to allow the longitudinally distal portion of the implant, typically including the tubular portion within which valve member **58** is disposed, to expand).

For some applications, delivery tool **89** (e.g., an extracorporeal control portion **110** thereof, disposed at a proximal portion of the delivery tool) is configured to ensure that the operator follows the particular sequence of movement of capsule-portions **92** and **94** described. For example, control portion **110** may comprise a series of locks **114**, each lock becoming unlockable only after the previous step in the sequence has been completed, the unlocking of the lock allowing the subsequent step in the sequence to be performed. For some such applications, control portion **110** may further comprise a series of controllers **112**, via which the operator causes the movement of a respective capsule-portion for each respective step. An example of how control portion **110** may be used, in accordance with some applications of the invention, is as follows:

(1) Operator moves distal capsule-portion **92** distally with respect to implant **20** by actuating a first controller **112a**. Once capsule-portion **92** has been moved a given distance, a first lock **114a** automatically becomes unlockable (and typically capsule-portion **92** is prevented from moving further than the given distance).

(2) Operator unlocks lock **114a**, and then moves proximal capsule-portion **94** proximally with respect to implant **20** by actuating a second controller **112b**. Once capsule-portion **94** has been moved a given distance, a second lock **114b** automatically becomes unlockable.

(3) Operator unlocks lock **114b**, and then moves distal capsule-portion **92** further distally with respect to implant **20** by again actuating first controller **112a**. Alternatively, control portion **110** may comprise a third controller (not shown) for this second movement of distal capsule-portion **92**.

It is to be noted that throughout this patent application (including in the specification and in the claims) the term “unlockable” means able to be unlocked (rather than meaning not lockable).

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Therefore, apparatus is provided, in accordance with some applications of the invention, the apparatus comprising an implant (e.g., implant 20) and a delivery tool (e.g., tool 89), the tool comprising:

- (1) a delivery capsule, disposed at a distal portion of the tool, the delivery capsule comprising a proximal capsule-portion 94 dimensioned to house a first part of the implant, and a distal capsule-portion 92 dimensioned to house a second part of the implant; and
- (2) an extracorporeal control portion 110, disposed at a proximal portion of the tool, the control portion comprising one or more controllers 112 and one or more locks 114, the controllers being operatively coupled to the proximal and distal capsule-portions such that the proximal and distal capsule-portions are movable with respect to the housed implant via actuation of the controllers,

wherein the controllers and the locks are mechanically cooperative such that:

- in a first state of the tool:
 - a first distal movement of the distal capsule-portion up to a first distance distally with respect to the implant is facilitated,
 - a second distal movement of the distal capsule-portion further distally with respect to the implant is inhibited,
 - a proximal movement of the proximal capsule-portion proximally with respect to the implant is inhibited, and
 - responsively to movement of the distal capsule-portion distally, the tool automatically unlocks a second state of the tool;
- in the second state of the tool:
 - the proximal movement of the proximal capsule-portion is facilitated,
 - the second distal movement of the distal capsule-portion remains inhibited, and
 - responsively to movement of the proximal capsule-portion proximally, the tool automatically unlocks a third state of the tool; and
- in the third state of the tool:
 - the second distal movement of the distal capsule-portion is facilitated.

For some applications, rather than having three states, tool 89 has two states. For clarity, these will be referred to as state A and state B. However, state A may be similar to the second state described hereinabove, *mutatis mutandis*. (State A may alternatively be called a primary state, and state B may alternatively be called a secondary state.) For applications in which tool 89 has two states, the tool typically has only one lock 114, the unlocking of which transitions the tool from state A to state B.

In state, A, it is possible to move distal capsule-portion 92 up to the first distance distally with respect to implant 20 (e.g., by actuating controller 112a), and to move proximal capsule-portion 94 proximally with respect to the implant (e.g., by actuating controller 112b), but it is not possible to move the distal capsule-portion further distally, which would deploy downstream region 33. Typically, in state A, it is also possible to return proximal capsule-portion 94 distally and distal capsule-portion 92 proximally. In state B, it is possible to move distal capsule-portion 92 further distally, thereby deploying downstream region 33. For some applications, lock 114 is not unlockable until distal capsule-portion 92 has been moved the first distance distally, and/or until proximal capsule-portion 94 has been moved proximally.

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Therefore, apparatus is provided, in accordance with some applications of the invention, the apparatus comprising an implant and a delivery tool, the tool comprising:

- a delivery capsule, disposed at a distal portion of the tool, the delivery capsule comprising a proximal capsule-portion dimensioned to house a first part of the implant, and a distal capsule-portion dimensioned to house a second part of the implant; and

an extracorporeal control portion, disposed at a proximal portion of the tool, the control portion comprising one or more controllers and one or more locks, the controllers being operatively coupled to the proximal and distal capsule-portions such that the proximal and distal capsule-portions are movable with respect to the implant via actuation of the controllers,

wherein the controllers and the locks are mechanically cooperative such that:

- in a state A of the tool: (i) a first distal movement of the distal capsule-portion up to a first distance distally with respect to the implant is facilitated, (ii) a second distal movement of the distal capsule-portion further distally with respect to the implant is inhibited, and (iii) a proximal movement of the proximal capsule-portion proximally with respect to the implant is facilitated; and

in a state B of the tool, the second distal movement of the distal capsule-portion is facilitated.

As shown in FIG. 1A, for some applications, in the expanded state of frame assembly 22, implant 20 defines a toroidal space 49 between flanges 54 and upstream support portion 40 (e.g., a space that is wider than distance d4). For example, space 49 may have a generally triangular cross-section. It is hypothesized that for some such applications, in addition to sandwiching tissue of the native valve between upstream support portion 40 and flanges 54 (e.g., the tips of the flanges), space 49 advantageously promotes tissue growth therewithin (e.g., between leaflet tissue and covering 23), which over time further secures implant 20 within the native valve.

Reference is now made to FIG. 5, which is a schematic illustration of a step in the implantation of implant 20, in accordance with some applications of the invention. Whereas FIGS. 4A-F show an implantation technique in which flanges 54 are expanded prior to upstream support portion 40, for some applications the upstream support portion is expanded prior to the flanges. FIG. 5 shows a step in such an application.

Reference is again made to FIGS. 2A-5. As noted hereinabove, implant 20 may be implanted by causing flanges 54 to radially protrude before causing upstream support portion 40 to radially protrude, or may be implanted by causing the upstream support portion to protrude before causing the flanges to protrude. For some applications, implant 20 is thereby configured to be deliverable in a downstream direction (e.g., transeptally, as shown, or transapically) or in an upstream direction (e.g., transapically or via the aortic valve). Thus, for some applications, an operating physician may decide which delivery route is preferable for a given application (e.g., for a given subject, and/or based on available equipment and/or expertise), and implant 20 is responsively prepared for the chosen delivery route (e.g., by loading the implant into an appropriate delivery tool).

It is to be noted that for some applications, downstream delivery of implant 20 may be performed by expanding flanges 54 first (e.g., as shown in FIGS. 4A-F) or by expanding upstream support portion 40 first (e.g., as shown in FIG. 5). Similarly, for some applications upstream deliv-

ery of implant **20** may be performed by upstream support portion **40** first, or by expanding flanges **54** first.

Reference is now made to FIG. **6**, which is a schematic illustration of implant **20**, in the state and position shown in FIG. **4D**, in accordance with some applications of the invention. For some applications, while implant **20** is in the state and position shown in FIG. **4D**, leaflets **12** of valve **10** are able to move, at least in part in response to beating of the heart. Frame (A) shows leaflets **12** during ventricular systole, and frame (B) shows the leaflets during ventricular diastole. For some such applications, blood is thereby able to flow from atrium **6** to ventricle **8**, between leaflets **12** and implant **20**. It is hypothesized that this advantageously facilitates a more relaxed implantation procedure, e.g., facilitating retaining of implant **20** in this state and position for a duration of greater than 8 minutes. During this time, imaging techniques may be used to verify the position of implant **20**, and/or positioning of leaflets **12** between upstream support portion **40** and flanges **54**.

Reference is made to FIGS. **7A-B** and **8A-B**, which are schematic illustrations of frame assemblies **122** and **222** of respective implants, in accordance with some applications of the invention. Except where noted otherwise, frame assemblies **122** and **222** are typically identical to frame assembly **22**, mutatis mutandis. Elements of frame assemblies **122** and **222** share the name of corresponding elements of frame assembly **22**. Additionally, except where noted otherwise, the implants to which frame assemblies **122** and **222** belong are similar to implant **20**, mutatis mutandis.

Frame assembly **122** comprises (i) a valve frame **130** that comprises a tubular portion **132** and an upstream support portion **140** that typically comprises a plurality of arms **146**, and (ii) an outer frame (e.g., a leg frame) **160** that circumscribes the valve frame, and comprises a plurality of legs **150** that each comprise a tissue-engaging flange **154**. Typically, outer frame **160** comprises a ring **166** to which legs **150** are coupled. Ring **166** is defined by a pattern of alternating peaks and troughs, the peaks being fixed to frame **130** at respective coupling points **152**, e.g., as described hereinabove for frame assembly **22**, mutatis mutandis.

Frame assembly **222** comprises (i) a valve frame **230** that comprises a tubular portion **232** and an upstream support portion **240** that typically comprises a plurality of arms **246**, and (ii) an outer frame (e.g., a leg frame) **260** that circumscribes the valve frame, and comprises a plurality of legs **250** that each comprise a tissue-engaging flange **254**. Typically, outer frame **260** comprises a ring **266** to which legs **250** are coupled. Ring **266** is defined by a pattern of alternating peaks and troughs, the peaks being fixed to frame **230** at respective coupling points **252**, e.g., as described hereinabove for frame assembly **22**, mutatis mutandis.

Whereas arms **46** of frame assembly **22** are shown as extending from upstream end **34** of tubular portion **32**, arms **146** and **246** of frame assemblies **122** and **222**, respectively, extend from sites further downstream. (This difference may also be made to frame assembly **22**, mutatis mutandis.) Tubular portions **32**, **132** and **232** are each defined by a repeating pattern of cells that extends around the central longitudinal axis. Typically, and as shown, tubular portions **32**, **132** and **232** are each defined by two stacked, tessellating rows of cells. In the expanded state of each tubular portion, these cells are typically narrower at their upstream and downstream extremities than midway between these extremities. For example, and as shown, the cells may be roughly diamond or astroid in shape. In frame assembly **22**, each arm **46** is attached to and extends from a site **35** that is at the upstream extremity of cells of the upstream row. In

contrast, in frame assemblies **122** and **222**, each arm **146** or **246** is attached to and extends from a site **135** (assembly **122**) or **235** (assembly **222**) that is at the connection between two adjacent cells of the upstream row (alternatively described as being at the upstream extremity of cells of the downstream row).

It is hypothesized by the inventors that this lower position of the arms, while maintaining the length of the lumen of the tubular portion, advantageously reduces the distance that the tubular portion (i.e., the downstream end thereof) extends into the ventricle of the subject, and thereby reduces a likelihood of inhibiting blood flow out of the ventricle through the left ventricular outflow tract. It is further hypothesized that this position of the arms reduces radial compression of the tubular portion by movement of the heart, due to greater rigidity of the tubular portion at sites **135** and **235** (which is supported by two adjacent cells) than at site **35** (which is supported by only one cell).

As shown, in the expanded state of frame assemblies **22**, **122** and **222**, the legs (**50**, **150** and **250**, respectively) are circumferentially staggered with the arms of the upstream support portion (**46**, **146** and **246**, respectively). This allows the legs to move in an upstream direction between the arms during expansion of the tubular portion (**32**, **132** and **232**, respectively), facilitating application of greater sandwiching force on tissue of the native valve. The lower position of the arms of assemblies **122** and **222** includes circumferentially shifting the position of the arms by the width of half a cell. In order to maintain the circumferential staggering of the arms and legs, rings **166** and **266** (and thereby legs **150** and **250**) are circumferentially shifted correspondingly. As a result, whereas the peaks of ring **66** generally align with connections between adjacent cells of the downstream row of cells of tubular portion **32** (and are fixed to these sites), the peaks of rings **166** and **266** are generally aligned midway between these sites (i.e., at spaces of the cellular structure of the tubular portion). Appendages **168** (for assembly **122**) or **268** (for assembly **222**) facilitate fixing of the peak with respect to the tubular structure.

For assembly **122**, appendages **168** are defined by valve frame **130** (e.g., by tubular portion **132** thereof) and extend (in a downstream direction) to the peaks of ring **166**, to which they are fixed. For example, each appendage **168** may define a valve-frame coupling element **131** that is fixed to a respective outer-frame coupling element **161** defined by outer frame **260**. Typically, appendages **168** extend from sites **135**. Typically, appendages **168** are integral with tubular portion **132** and/or in-plane with the tubular portion (e.g., are part of its tubular shape).

For assembly **222**, appendages **268** are defined by outer frame **260**, and extend (e.g., in an upstream direction) from the peaks of ring **266**. Typically, appendages **268** extend to sites **235**, to which they are fixed. For example, each appendage **268** may define an outer-frame coupling element **261** that is fixed to a respective valve-frame coupling element **231** defined by valve frame **230** (e.g., by tubular portion **232** thereof). Typically, appendages **268** are integral with outer frame **260** and/or in-plane with adjacent portions of outer frame **260**, such as ring **266**.

Therefore, frame assembly **122** defines a hub at site **135**, and frame assembly **222** defines a hub at site **235**. For some applications, apparatus therefore comprises:

- a plurality of prosthetic valve leaflets; and
- a frame assembly, comprising:

- a tubular portion (**132** or **232**) defined by a repeating pattern of cells, the tubular portion extending circumferentially around longitudinal axis ax1 so as to

define a longitudinal lumen, the prosthetic valve leaflets coupled to the inner frame and disposed within the lumen;

an outer frame (160 or 260), comprising a plurality of legs (150 or 250), distributed circumferentially around the tubular portion, each leg having a tissue-engaging flange (154 or 254);

an upstream support portion (140 or 240) that comprises a plurality of arms (146 or 246) that extend radially outward from the tubular portion; and

a plurality of appendages (168 or 268), each having a first end that defines a coupling element (161 or 261) via which the tubular portion is coupled to the outer frame, and a second end;

wherein the frame assembly defines a plurality of hubs (135 or 235), distributed circumferentially around the longitudinal axis on a plane that is transverse to longitudinal axis ax1, each hub defined by convergence and connection of, (i) two adjacent cells of the tubular portion, (ii) an arm of the plurality of arms, and (iii) an appendage of the plurality of appendages.

Reference is made to FIGS. 9A-C, which are schematic illustrations of an implant 320 comprising a frame assembly 322, in accordance with some applications of the invention. Except where noted otherwise, frame assembly 322 is identical to frame assembly 122, and implant 300 is identical to the implant to which frame assembly 122 belongs, *mutatis mutandis*. FIG. 9A is a side-view of implant 320, and FIG. 9B is an isometric bottom-view of the implant.

Frame assembly 122 comprises (i) a valve frame 330 that comprises a tubular portion 332 and an upstream support portion 340 that typically comprises a plurality of arms 346, and (ii) an outer frame (e.g., a leg frame) 360 that circumscribes the valve frame, and comprises a plurality of legs 350 that each comprise a tissue-engaging flange 354. Typically, outer frame 360 comprises a ring 366 to which legs 350 are coupled. Ring 366 is defined by a pattern of alternating peaks and troughs, the peaks being fixed to frame 330 at respective coupling points 352, e.g., as described hereinabove for frame assembly 22 and/or frame assembly 122, *mutatis mutandis*.

Frame assembly 322 comprises an annular upstream support portion 340 that has an inner portion 342 that extends radially outward from the upstream portion (e.g., the upstream end) of tubular portion 332. Upstream support portion 340 further comprises one or more fabric pockets 344 disposed circumferentially around inner portion 342, each pocket of the one or more pockets having an opening that faces a downstream direction (i.e., generally toward the downstream end of implant 320). In the figures, upstream support portion 340 has a single toroidal pocket 344 that extends circumferentially around inner portion 342.

Typically, a covering 323 (e.g., similar to covering 23, described hereinabove, *mutatis mutandis*) is disposed over arms 346, thereby forming pocket 344. Further typically, arms 346 are shaped to form pocket 344 from covering 323. For example, and as shown, arms 346 may curve to form a hook-shape.

For some applications, portion 340 has a plurality of separate pockets 344, e.g., separated at arms 346. For some such applications, covering 323 is loosely-fitted (e.g., baggy) between radially-outward parts of arms 346, e.g., compared to inner portion 342, in which the covering is more closely-fitted between radially-inward parts of the arms.

FIG. 9C shows implant 320 implanted at native valve 10. Pocket 344 is typically shaped and arranged to billow in

response to perivalvular flow 302 of blood in an upstream direction. If ventricular systole forces blood in ventricle 8 between implant 320 and native valve 10, that blood inflates pocket 344 and presses it (e.g., covering 323 and/or the radially-outward part of arm 346) against tissue of atrium 6 (e.g., against the atrial wall), thereby increasing sealing responsively. It is hypothesized by the inventors that the shape and orientation of pocket 344 (e.g., the hook-shape of arms 346) facilitates this pressing radially-outward in response to the pocket's receipt of upstream-flowing blood.

Pocket(s) 344 may be used in combination with any of the implants described herein, *mutatis mutandis*.

Reference is again made to FIGS. 1A-9C. It is to be noted that unless specifically stated otherwise, the term "radially outward" (e.g., used to describe upstream support portion 40 and flanges 54) means portions of the element are disposed progressively further outward from a central point (such as longitudinal axis ax1 or tubular portion 32), but does not necessarily mean disposed at 90 degrees with respect to longitudinal axis ax1. For example, flanges 54 may extend radially outward at 90 degrees with respect to longitudinal axis ax1, but may alternatively extend radially outward at a shallower angle with respect to the longitudinal axis.

It will be appreciated by persons skilled in the art that the present invention is not limited to what has been particularly shown and described hereinabove. Rather, the scope of the present invention includes both combinations and subcombinations of the various features described hereinabove, as well as variations and modifications thereof that are not in the prior art, which would occur to persons skilled in the art upon reading the foregoing description.

The invention claimed is:

1. Apparatus for use with a native valve that is disposed between an atrium and a ventricle of a heart of a subject, the apparatus comprising:

a valve frame, comprising:

a tubular portion that circumscribes a longitudinal axis of the valve frame so as to define a lumen along the axis, the tubular portion defining a plurality of valve-frame coupling elements disposed circumferentially around the longitudinal axis; and

a plurality of arms, extending radially outward from an upstream portion of the tubular portion to define an arm span, and configured to engage tissue in the atrium;

a plurality of prosthetic leaflets, coupled to the frame, disposed within the lumen, and arranged to provide unidirectional flow of blood from an upstream end of the lumen to a downstream end of the lumen; and

an outer frame:

coupled to the valve frame, and

comprising a ring that circumscribes the tubular portion, and a plurality of flanges that extend radially outward from the ring to define a flange span, and are configured to

engage tissue of the ventricle,

wherein:

each arm of the plurality of arms defines a rigid portion and a flexible portion that is disposed radially outward from the rigid portion, and is more flexible than the rigid portion, and the majority of the flexible portion is disposed further radially outward than the flange span.

2. The apparatus according to claim 1, wherein the apparatus is configured such that the rigid portion provides resistance against axial movement of the apparatus while the flexible portion facilitates conformation of each arm of the

plurality of arms to anatomy of the atrium and/or lateral movement of the apparatus within the native valve.

3. The apparatus according to claim 1, wherein each arm of the plurality of arms defines a serpentine shape in the flexible portion. 5

4. The apparatus according to claim 1, wherein each arm of the plurality of arms defines a plurality of holes in the flexible portion.

5. The apparatus according to claim 1, wherein less than 90 percent of the flexible portion is disposed further radially outward than the flange span. 10

6. The apparatus according to claim 1, wherein more than 60 percent of the flexible portion is disposed further radially outward than the flange span.

7. The apparatus according to claim 1, wherein, for each arm of the plurality of arms, the flexible portion covers an overall surface area including (i) the area occupied by material from which the arm is formed, and (ii) spaces, and the material from which the arm is formed occupies less than 80 percent of the overall surface area. 15 20

8. The apparatus according to claim 7, wherein the material from which the arm is formed occupies more than 20 percent of the overall surface area.

9. The apparatus according to claim 7, wherein the material from which the arm is formed occupies less than 70 percent of the overall surface area. 25

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